



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	248	Media
TechCrunch AI	13	Media
HackerNews	7	Community
The Verge AI	7	Media
MIT Tech Review	1	Media

VERIFIED INTELLIGENCE — 3 stories

VERI

Show HN: Geiger - See every AI agent on your machine and what it can touch

HackerNews

+1 source

90%

HN 45

Hi HN, I am Dimitris. This is a company OS that I built and use to run my business and anyone can install it and self host it for free. Think of it as Claude Code, Cowork and the cloud sessions in one self hosted application. It is a Multi-tenant application by design where many...

<https://github.com/Atomburstofficial/geiger>

VERI

Apple's new iPhone camera mode promises to prove your photo isn't AI

The Verge AI

+1 source

80%

Apple is launching a new way to prove that the picture you took isn't manipulated by AI. A new feature, called "Reference Image," will arrive with the iPhone 18 Pro lineup later this month and is supposed to use the device's new camera sensor to "sign every pixel it sees." The...

<https://www.theverge.com/tech/992766/apple-iphone-18-pro-reference-image>

VERI

Worried Anthropic researchers warn that AI could kill all humans;

The Verge AI

+1 source

80%

A senior Anthropic safety researcher has said there is more than a 10 percent chance artificial intelligence "could kill all humans" by the end of the decade, just hours after a colleague resigned over fears the AI lab and its rivals are carelessly racing to build "superhuman..."

<https://www.theverge.com/ai-artificial-intelligence/991927/anthropic-ai-kill-all-humans>

CONFIRMED SIGNALS — 270 stories

CONF

Show HN: Give your AI agent on-screen guides that show users where to click

HackerNews

50%

HN 18

Hey HN. I'm Christian, one of the founders of Frigade (YC W23). I've noticed that a lot of in-app AI agents struggle to actually understand the products they exist in. For instance, let's say a user asks an agent how to do something in a given SaaS product. In an ideal case,...

<https://news.ycombinator.com/item?id=49627872>

CONF

Show HN: Castforge - run Claude Code, Codex and Gemini as one dev team

HackerNews

50%

HN 5

No summary — see link for full article.

<https://castforge.ai/>

CONF

Show HN: Locksmith: Store Maven Secrets in the macOS Keychain (Or a Unix Socket)

HackerNews

50%

HN 3

Hello HN, In the age of LLMs sucking in random secrets, sandbox escapes, and RCE orbital bombardment, I thought it'd be a good idea to create a Maven Extension that defers to the MacOS keychain for resolving secrets. In general, it's a good idea to not have any secrets...

<https://github.com/exabrial/locksmith>

CONF

Show HN: Stop Scattering AI Resources Across Your Project: Meet Bear. Ctxpm

HackerNews

50%

HN 3

No summary — see link for full article.

<https://bear.best/en/blog/bear-ctxpm-context-package-manager/>

CONF

Show HN: TrailerPark, a place just to talk about film trailers

HackerNews 50% HN 2

I've watched and re-watched thousands of movie trailers. Some trailers are better than the movie. So I built a place I can watch, revisit and discuss some of my favorites. It's very stupid but I think that's half the appeal: <https://www.queuebrick.com/trailerpark/>
<https://news.ycombinator.com/item?id=49628138>

CONF

Show HN: Emailclaw-Email lovers' AI agent that creates scheduled task in 5 steps

HackerNews 50% HN 2

No summary — see link for full article.

<https://github.com/emailclaw/emailclaw>

0 CONF

AI research startup Listen Labs scrubbed a \$1.5B funding round for Salesforce talks

TechCrunch AI 50%

Listen Labs walked away from a signed Series C term sheet from Menlo Ventures, sources say.

<https://techcrunch.com/2026/09/09/ai-research-startup-listen-labs-scrubbed-a-1-5b-fundi...>

1 CONF

OpenAI adds a prominent AI doomer to its board of directors

TechCrunch AI 50%

Paul Christiano, an influential AI researcher focused on alignment, is joining the OpenAI Foundation as a member of its board.

<https://techcrunch.com/2026/09/09/openai-adds-a-prominent-ai-doomer-to-its-board-of-dir...>

2 CONF

Apple Watch's new AI features are normalizing the idea that technology is always listening

TechCrunch AI 50%

Apple says its new watches won't save raw audio, but features that can transcribe recent speech and summarize ambient conversations raise new questions about consent, privacy, and how people behave when they know they could always be recorded.

<https://techcrunch.com/2026/09/09/apple-watchs-new-ai-features-are-normalizing-the-idea...>

3 CONF

The hinge for Apple's new foldable phone was built with AI

TechCrunch AI 50%

Apple says it used AI and 3D printing in the manufacturing process for its long-awaited foldable phone.

<https://techcrunch.com/2026/09/09/the-hinge-for-apples-new-foldable-phone-was-built-wit...>

4 CONF

Apple CEO John Ternus says the best AI device is still the iPhone

TechCrunch AI 50%

The company also argued that its on-device models offer consumers more privacy.

<https://techcrunch.com/2026/09/09/apple-ceo-john-ternus-says-the-best-ai-device-is-stil...>

5 CONF

ControlAI's Connor Leahy on why superintelligence is 'not a weapon, it's an adversary'

TechCrunch AI 50%

AI companies have been talking about superintelligent AI like it's inevitable, but recent safety incidents like OpenAI's Hugging Face breach are demonstrating the potential dangers of deploying AI systems that are more capable than humans. So what happens when we can't reliably...

<https://techcrunch.com/podcast/controlais-connor-leahy-on-why-superintelligence-is-not-...>

6 CONF

Viral AI assistant Instinct now has its own email address

TechCrunch AI 50%

Instinct's new email feature lets the AI agent create and manage accounts, contact businesses, handle support requests, and do more on users' behalf.

<https://techcrunch.com/2026/09/09/viral-ai-assistant-instinct-now-has-its-own-email-add...>

7 CONF

Shipt becomes the latest delivery app with an AI shopping assistant

TechCrunch AI 50%

Users can ask the assistant to do things like "Create a cart for my Saturday tailgate for 25 people and include some brunch items," or "Build a cart for easy school lunches and after-school snacks," Shipt says.

<https://techcrunch.com/2026/09/09/shipt-becomes-the-latest-delivery-app-with-an-ai-shop...>

8 CONF

AI spend per employee slumped at top firms in August — summer doldrums or a warning sign?

TechCrunch AI 50%

Falling token costs, cheaper models, and less spend per employee — AI adoption isn't playing out the way hyperscalers hoped.

<https://techcrunch.com/2026/09/09/ai-spend-per-employee-slumped-at-top-firms-in-august-...>

9 CONF

Sequoia doubles down on Cymphony as AI agents create new enterprise security risks

TechCrunch AI 50%

Cymphony gives security teams a single view of employees, AI agents, and other nonhuman identities, including the systems and sensitive data they can access.

<https://techcrunch.com/2026/09/09/sequoia-doubles-down-on-cymphony-as-ai-agents-create-...>

10 CONF

Suno replaces its AI models with a new one trained on licensed music as copyright suits pile up

TechCrunch AI 50%

As it grapples with a bevy of lawsuits, Suno said its new model, Suno v6, is not trained using music it used to train previous versions of the AI model.

<https://techcrunch.com/2026/09/09/suno-replaces-its-ai-models-with-a-new-one-trained-on-...>

1 CONF

Suno releases its first AI music model made with record industry help

The Verge AI 50%

Suno's new v6 AI music model is its first made with support from the record industry. Suno's Jack Brody told The Verge that v6 was "trained from the ground up, with a new set of data that does not include the same data that our previous models were trained on." The data includes...

<https://www.theverge.com/ai-artificial-intelligence/991977/suno-releases-its-first-ai-m...>

2 CONF

OpenAI's sly mathematical breakthrough sends a chill through academia

The Verge AI 50%

OpenAI's announcement Tuesday that it has solved one of mathematics' legendary Millennium Prize problems should have been a moment of triumph. The result is both an undeniable achievement and a striking demonstration of just how rapidly AI is transforming mathematics. But before...

<https://www.theverge.com/ai-artificial-intelligence/992953/openai-math-millennium-prize-...>

3 **CONF****Read the Apple document explaining how new listening features still protect your privacy****The Verge AI** 50%

At Wednesday's iPhone Duo launch event, Apple announced a handful of new Siri AI Audio Intelligence features, including Siri Recap, Live Rewind, Sound Recognition, and Music Recognition. Alongside its announcement, Apple released a document laying out how it plans to balance AI...

<https://www.theverge.com/tech/992919/apple-siri-ai-audio-intelligence-privacy>
4 **CONF****Microsoft has new AI privacy rules for schools****The Verge AI** 50%

Microsoft agreed to a set of safety and privacy principles for AI in schools a week after two major school systems announced a ban on student-facing AI. In a new agreement with the American Federation of Teachers (AFT), the second-largest teachers union in the US, and its New...

<https://www.theverge.com/policy/992359/microsoft-aft-schools-ai-privacy>
5 **CONF****Amazon Prime Video's new AI tech matches lips to dubbed audio****The Verge AI** 50%

Amazon's Prime Video is launching a new AI-powered feature that lines up an actor's mouth with "human-dubbed" audio. The feature is only available with the English dub of the German series Maxton Hall for now, but Prime Video plans to expand it to "additional titles" in the...

<https://www.theverge.com/tech/991809/amazon-prime-video-ai-lip-sync-dubbing>
6 **CONF****When Does Memory Help? A Cost-Aware Evaluation of Long-Term Memory in Tool-Using LLM Agents****ArXiv CS.AI** 50%

arXiv:2609.05441v1 Announce Type: new Abstract: Long-term memory for LLM agents is evaluated today by conversational recall benchmarks (LoCoMo, LongMemEval), which measure question answering over dialogue history, not whether remembered facts change what a tool-using agent does....

<https://arxiv.org/abs/2609.05441>
7 **CONF****Spike-Aware INT8 Execution for Spiking Language Models on Commodity CPUs****ArXiv CS.AI** 50%

arXiv:2606.03026v2 Announce Type: replace-cross Abstract: Binary spike activations allow a language-model runtime to read only active weight columns and replace multiplications by weight sums. We implement this execution strategy in C++ for an 874M-parameter spike-gated language...

<https://arxiv.org/abs/2606.03026>
8 **CONF****RAPID: Reliability-Aware Pair Importance Distillation****ArXiv CS.AI** 50%

arXiv:2609.05481v1 Announce Type: new Abstract: Inter example relational distillation transfers a teacher's representation geometry by matching relations among examples within a mini batch. Computing all pairs has quadratic complexity in the batch size, whereas uniform...

<https://arxiv.org/abs/2609.05481>

9 CONF

xDailyBench: Benchmarking LLMs on Professional Consultation for Real-Life Problems

ArXiv CS.AI 50%

arXiv:2609.07784v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly used for everyday assistance, yet existing benchmarks only partially reflect the requests users naturally make in practice. Real-world requests are often open-ended, casually specified,...

<https://arxiv.org/abs/2609.07784>

0 CONF

Reasoning-Aware Compression: Identifying and Protecting Vulnerable Reasoning Circuits for Energy-Efficient LLM Deployment

ArXiv CS.AI 50%

arXiv:2609.05512v1 Announce Type: new Abstract: Large Reasoning Models (LRMs) impose substantial energy costs during deployment, yet current compression methods apply uniform quantization across all components, risking damage to critical reasoning circuits. We present a...

<https://arxiv.org/abs/2609.05512>

1 CONF

The Failure Happens Before the Drift: The Social Dynamics of Values in LLM Agent Societies

ArXiv CS.AI 50%

arXiv:2609.05514v1 Announce Type: new Abstract: Large Language Model (LLM)-based agents are increasingly used as proxies for human participants in social science research, yet it remains unclear whether they can faithfully simulate diverse and conflicting human value systems. We...

<https://arxiv.org/abs/2609.05514>

2 CONF

EdgeMem: LLM-Free Agent Memory Construction and Retrieval via Evidence-Preserving Multi-Anchor Hypergraph

ArXiv CS.AI 50%

arXiv:2609.05553v1 Announce Type: new Abstract: Agent memory allows LLM agents to use earlier interactions when answering new queries. Existing methods often compress interaction histories into summaries or other LLM-generated representations. Repeated generation adds cost and...

<https://arxiv.org/abs/2609.05553>

3 CONF

Planning and Scheduling Business Processes under Control-Flow Uncertainty

ArXiv CS.AI 50%

arXiv:2609.05578v1 Announce Type: new Abstract: Scheduling activities in business processes can improve efficiency (e.g., reduce makespan), but is challenging because the exact sequence of activities required to complete a case is often uncertain due to decisions based on data...

<https://arxiv.org/abs/2609.05578>

4 CONF

The convergent laboratory: when AI reasoning, autonomous experiments, high performance and quantum computing reshape chemistry

ArXiv CS.AI 50%

arXiv:2609.05643v1 Announce Type: new Abstract: This Comment emerges from TPC26 (<https://tpc26.org>), a conference convening leaders from academia, national laboratories, and industry who are reshaping materials science discovery. The meeting explored how AI, autonomous agents,...

<https://arxiv.org/abs/2609.05643>

5 **CONF****What LLM Trading Agents Actually Do in Production: A Six-Month, Population-Scale Record from Two Fleets****ArXiv CS.AI** **50%**

arXiv:2609.05663v1 Announce Type: new Abstract: We present a continuous, population-scale measurement record of autonomous language-model trading agents operating in production across two systems with one design lineage: DX Terminal Pro (3,505 user-funded vaults trading real ETH...

<https://arxiv.org/abs/2609.05663>

6 **CONF****Decoupled Temporal Encoding for Generative Recommendation****ArXiv CS.AI** **50%**

arXiv:2608.16274v2 Announce Type: replace-cross Abstract: Positional encoding is a fundamental component of Transformer-based generative recommendation models, where user histories are modeled as autoregressive item sequences. Most positional encoding methods are inherited from...

<https://arxiv.org/abs/2608.16274>

7 **CONF****Audio-Maestro: Enhancing Large Audio-Language Models with Tool-Augmented Reasoning****ArXiv CS.AI** **50%**

arXiv:2510.11454v2 Announce Type: replace-cross Abstract: Recent advancements in large multimodal models (LMMs) have shown strong capabilities in audio understanding. However, most systems rely solely on end-to-end reasoning, limiting interpretability and accuracy for tasks that...

<https://arxiv.org/abs/2510.11454>

8 **CONF****Beyond Prompts: Measuring and Optimizing LLM Tool-Agent Harnesses****ArXiv CS.AI** **50%**

arXiv:2609.05736v2 Announce Type: new Abstract: LLM tool agents can be improved without retraining by modifying the runtime harness around a fixed model: prompts, tool interfaces, middleware, state handling, and recovery logic. We study this setting as resource-bounded harness...

<https://arxiv.org/abs/2609.05736>

9 **CONF****SchemeArena: Factorized Stress Testing of Scheming in LLM Agents****ArXiv CS.AI** **50%**

arXiv:2609.08126v1 Announce Type: new Abstract: We study scheming in LLM agents, in which agents covertly pursue misaligned goals. Our focus is to understand how scheming arises from the interaction of key factors, such as instrumental goals, environmental affordances, oversight...

<https://arxiv.org/abs/2609.08126>

10 **CONF****Inference-Time Nash Alignment****ArXiv CS.AI** **50%**

arXiv:2609.08082v1 Announce Type: new Abstract: Preference-based fine-tuning methods such as RLHF and DPO require substantial compute and large preference datasets. They also need direct access to the model parameters which are not provided by many state-of-the-art models....

<https://arxiv.org/abs/2609.08082>

1 **CONF****DI-Bench: Systematically Generating In-Domain Data Intelligence Benchmarks for Enterprise Agents****ArXiv CS.AI** **50%**

arXiv:2609.05776v1 Announce Type: new Abstract: Evaluating enterprise agents on domain-specific benchmarks is critical, yet public benchmarks rarely evaluate whether agents can integrate business knowledge with analytical computation, and constructing such benchmarks manually is...

<https://arxiv.org/abs/2609.05776>

2 **CONF****In-Place Instruction Following in Diffusion Language Models****ArXiv CS.AI** **50%**

arXiv:2609.07160v1 Announce Type: cross Abstract: Diffusion Large Language Models (dLLMs) generate text via bidirectional iterative denoising, naturally supporting user-specified constraints anchored at arbitrary output positions, a paradigm known as In-place Prompting (IPP). We...

<https://arxiv.org/abs/2609.07160>

3 **CONF****More Than Mimicking Reviewers: Evaluating LLMs for Pre-Submission Peer Review****ArXiv CS.AI** **50%**

arXiv:2609.05788v1 Announce Type: new Abstract: Peer-review feedback often arrives too late for authors to make meaningful revisions. We study an author-facing LLM system that moves part of this stress test before submission: it generates a broad pool of atomic concerns and...

<https://arxiv.org/abs/2609.05788>

4 **CONF****Spillover-Aware Multi-Value Steering for Pluralistic LLM Alignment****ArXiv CS.AI** **50%**

arXiv:2609.05800v1 Announce Type: new Abstract: Activation steering controls LLM behavior at inference time by adding learned directions to hidden states, but existing methods handle one concept at a time. Pluralistic alignment, where different stakeholders need different value...

<https://arxiv.org/abs/2609.05800>

5 **CONF****Agentic BAIM-LLM Evaluation (ABLE): Benchmarking LLM Use of Protein Design Tools****ArXiv CS.AI** **50%**

arXiv:2609.05818v1 Announce Type: new Abstract: We introduce ABLE, a benchmark for evaluating LLM agents' ability to use biological AI models (BAIMs), such as ProteinMPNN and AlphaFold3, in dual-use protein design workflows. ABLE assesses agent performance through a set of tasks...

<https://arxiv.org/abs/2609.05818>

6 **CONF****Multimodal Resource-Exhaustion Attacks on Vision-Language Models via Joint Pixel-Prompt Optimization****ArXiv CS.AI** **50%**

arXiv:2609.05889v1 Announce Type: new Abstract: Resource-exhaustion attacks against autoregressive vision-language models (VLMs) typically assume unimodal threat models, treating the image branch as the primary optimization surface while holding user-visible prompts fixed. Even...

<https://arxiv.org/abs/2609.05889>

7 **CONF****Explaining AI Agents Through Execution Traces**

ArXiv CS.AI 50%

arXiv:2609.06063v1 Announce Type: new Abstract: AI Agents are increasingly deployed in real-world settings, where they interact with external tools and make sequential decisions with limited human oversight. This creates a pressing need for reliable and auditable explanations of...

<https://arxiv.org/abs/2609.06063>

8 **CONF****Emo-DVS: A Multimodal Benchmark for Privacy-Aware Emotion Recognition with Event Cameras**

ArXiv CS.AI 50%

arXiv:2609.06928v1 Announce Type: cross Abstract: Emotion analysis is a fundamental task in computer vision, but its practical deployment remains constrained by the privacy risks inherent to conventional RGB cameras. Bio-inspired event cameras present a promising hardware-level...

<https://arxiv.org/abs/2609.06928>

9 **CONF****Attribution in Scientific Literature: New Benchmark and Methods**

ArXiv CS.AI 50%

arXiv:2405.02228v4 Announce Type: replace-cross Abstract: Large language models (LLMs) increasingly generate citation-backed responses, yet citation hallucination remains a major challenge for trustworthy scientific information access. We introduce REASONS, a benchmark of 12,723...

<https://arxiv.org/abs/2405.02228>

10 **CONF****Building Trustworthy Graph-Agentic RAG for Social Good: Architectures, Failure Propagation, and Assurance by Construction**

ArXiv CS.AI 50%

arXiv:2609.06391v1 Announce Type: new Abstract: Graph-agentic retrieval-augmented generation combines structured evidence with adaptive controllers that can plan retrieval, traverse relations, verify intermediate claims, delegate subtasks, and use tools. This combination is...

<https://arxiv.org/abs/2609.06391>

11 **CONF****When Intelligence Becomes Agency: A Theory of Governed, Proactive Agency for Symbiotic AI Systems**

ArXiv CS.AI 50%

arXiv:2609.07741v1 Announce Type: new Abstract: Persistent AI assistants are intended to extend human attention, memory, and coordination across changing digital and physical environments. To be truly useful they must do more than just act when asked. They must decide on their...

<https://arxiv.org/abs/2609.07741>

12 **CONF****A Translational Note on AI Safety Evaluation**

ArXiv CS.AI 50%

arXiv:2609.06573v1 Announce Type: new Abstract: Recent studies report that automated red-teaming finds more vulnerabilities, at lower cost, than human red-teaming on standard AI safety benchmarks, and some read this as evidence that human evaluators are becoming dispensable. The...

<https://arxiv.org/abs/2609.06573>

3 CONF

WorldAgen: Unified State-Action Prediction with Test-Time World Model Training

ArXiv CS.AI 50%

arXiv:2609.08162v1 Announce Type: new Abstract: How can vision-language-action (VLA) models adapt to new environments where world dynamics shift? While recent research has combined world modeling and action prediction to improve VLA performance, existing methods largely rely on...

<https://arxiv.org/abs/2609.08162>

4 CONF

We Built a Mirror and Mistook It for a Mind: Causal Liability and the Fallacy of AI Consciousness

ArXiv CS.AI 50%

arXiv:2609.06715v1 Announce Type: new Abstract: The contemporary debate over machine consciousness begins from a concealed assumption: that the object called "AI" already constitutes the kind of entity to which consciousness could belong. This paper challenges that assumption by...

<https://arxiv.org/abs/2609.06715>

5 CONF

Monte Carlo-Based Ex-Ante Assessment of the Green Benefits of an AI-Driven Smart Agriculture Platform in Hainan

ArXiv CS.AI 50%

arXiv:2609.06737v1 Announce Type: new Abstract: Smart agriculture platforms are widely regarded as key carriers for implementing China's pesticide and fertilizer reduction, water-saving and carbon-reduction agendas, yet a unified quantitative framework for assessing their green...

<https://arxiv.org/abs/2609.06737>

6 CONF

AURA-Eval: Evaluation Framework for Acting Under Risk Awareness in LLM Agent Trajectories

ArXiv CS.AI 50%

arXiv:2609.06783v1 Announce Type: cross Abstract: LLM agents operate in workflows where unsafe actions can have real consequences. Existing safety evaluations often reduce behavior to a single score, obscuring risk recognition, pre-action detection, and safe task completion when...

<https://arxiv.org/abs/2609.06783>

7 CONF

When and Why LLM Causal Priors Help: Closed-Loop Prior Selection for Amortized Causal Inference

ArXiv CS.AI 50%

arXiv:2609.06941v1 Announce Type: new Abstract: Causal effect estimation asks how an outcome would change under an intervention, and medicine, economics, and public policy all treat it as a foundational task. Prior-data fitted networks (PFNs) amortize the task: a model trained...

<https://arxiv.org/abs/2609.06941>

8 CONF

LOBERT: Generative AI Foundation Model for Limit Order Book Messages

ArXiv CS.AI 50%

arXiv:2511.12563v2 Announce Type: replace Abstract: Modeling the dynamics of financial Limit Order Books (LOB) at the message level is challenging due to irregular event timing, rapid regime shifts, and the reactions of high-frequency traders to visible order flow. Previous LOB...

<https://arxiv.org/abs/2511.12563>

9 CONF

Diffusion models for eye-gaze trajectory generation using position and velocity representations

ArXiv CS.AI 50%

arXiv:2609.05522v1 Announce Type: cross Abstract: Eye-tracking data are expensive to collect, requiring specialized hardware and controlled laboratory conditions, and difficult to share because of privacy constraints. We address this using two complementary denoising diffusion...

<https://arxiv.org/abs/2609.05522>

0 CONF

Beyond Sparse Rewards: A New Benchmark and Structure-Aware Graph Alignment for Micro-Drama Understanding

ArXiv CS.AI 50%

arXiv:2609.07107v1 Announce Type: new Abstract: Micro-dramas, characterized by ultra-short durations and hyper-dense storylines, pose unique challenges for video understanding that conventional benchmarks fail to address. To bridge this gap, we introduce M-Drama, the first...

<https://arxiv.org/abs/2609.07107>

1 CONF

Encoded Early, Used Late: Where Transformers Begin to Act on an Inferred Partner's Expertise

ArXiv CS.AI 50%

arXiv:2609.07139v1 Announce Type: new Abstract: A transformer can make an attribute linearly decodable in its residual stream at a depth where that attribute does not yet influence the output. This gap between where information is readable and where it is used has been shown for...

<https://arxiv.org/abs/2609.07139>

2 CONF

Foundation Models for Generalizable Semantic and Goal-Oriented Communication

ArXiv CS.AI 50%

arXiv:2609.07853v1 Announce Type: cross Abstract: Semantic and goal-oriented communication is increasingly studied for 6G, but generalization beyond seen data remains a key weakness under tight rate budgets. Many existing systems overfit their training data and degrade sharply...

<https://arxiv.org/abs/2609.07853>

3 CONF

Segment Any Motion with Radar: Robust Multimodal Moving-Object Segmentation and Tracking

ArXiv CS.AI 50%

arXiv:2609.08346v1 Announce Type: cross Abstract: Moving-object perception must decide which image regions correspond to real motion and keep every instance identified over time. Methods that read motion from appearance, optical flow, or estimated trajectories lose that evidence...

<https://arxiv.org/abs/2609.08346>

4 CONF

Unraveling the Real Working Mechanism and Inherent Flaws of GAE: A Method for Interpreting Transformer Processes from an Economic Perspective

ArXiv CS.AI 50%

arXiv:2609.07213v1 Announce Type: new Abstract: We observe a phenomenon that current algorithmic research in the field of explainable artificial intelligence primarily pursues better performance on several proxy metrics. On the one hand, these proxy metrics themselves are more...

<https://arxiv.org/abs/2609.07213>

5 CONF

Alpha-R1: Alpha Screening with LLM Reasoning via Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2512.23515v2 Announce Type: replace-cross Abstract: Signal decay and regime shifts pose recurring challenges for data-driven investment strategies in non-stationary markets, where conventional time-series and machine learning approaches often struggle to generalize beyond...

<https://arxiv.org/abs/2512.23515>

6 CONF

Procedural Graphs: Self-Evolving Execution Structures for LLM Agents

ArXiv CS.AI 50%

arXiv:2609.09153v1 Announce Type: new Abstract: Large language models are increasingly deployed as agents that plan over long horizons and act through external tools. Most agents select actions through unconstrained generation over an accumulating history, leaving implicit the...

<https://arxiv.org/abs/2609.09153>

7 CONF

Weakly supervised neural network: segmentation of complex structures in X-ray microCT

ArXiv CS.AI 50%

arXiv:2609.07313v1 Announce Type: new Abstract: Segmentation of complex structures in X-ray tomographic data is a fundamental task in biomedical research, but it often requires large amounts of precisely annotated data, making fully supervised approaches costly and difficult to...

<https://arxiv.org/abs/2609.07313>

8 CONF

DART: Distributional Adversarial Recurrent Training for Algorithm Learning

ArXiv CS.AI 50%

arXiv:2609.05988v1 Announce Type: cross Abstract: Recurrent reasoning models (RRMs) can solve structured problems, achieving easy-to-hard generalization through iterative computation in hidden space. These models are typically trained with instance-level supervision, which...

<https://arxiv.org/abs/2609.05988>

9 CONF

Human-like moral judgments conceal divergent motive attributions in large language models

ArXiv CS.AI 50%

arXiv:2609.07353v1 Announce Type: new Abstract: Large language models (LLMs) are used to simulate human participants in psychological research. We asked whether LLMs that reproduce human evaluations of a whistleblower's moral character also reproduce the motive attributions that...

<https://arxiv.org/abs/2609.07353>

0 CONF

RAFM-SER++: A Lightweight Multimodal Emotion Recognition Framework for Real-Time Behavioral Monitoring in Surveillance Systems

ArXiv CS.AI 50%

arXiv:2609.07409v1 Announce Type: new Abstract: Recent multimodal Speech Emotion Recognition (SER) systems achieve high accuracy through interaction-heavy cross-modal transformers, but their computational cost limits deployment in latency-sensitive and resource-constrained...

<https://arxiv.org/abs/2609.07409>

1 **CONF****CIT-CAD: Constraint Intent Tree-based CAD Code Generation and Verification****ArXiv CS.AI 50%**

arXiv:2609.07434v1 Announce Type: new Abstract: Natural-language Computer-Aided Design (CAD) code generation aims to turn design intent into executable and editable parametric programs. Large language models (LLMs) make this goal increasingly practical, but useful systems must...

<https://arxiv.org/abs/2609.07434>

2 **CONF****Quantile-Led Feature Extraction for Multi-Horizon Predictive Maintenance in Industrial Manufacturing Systems****ArXiv CS.AI 50%**

arXiv:2609.07533v1 Announce Type: new Abstract: In data-driven predictive maintenance (PdM), feature extraction is usually treated as fixed preprocessing: a descriptor set is chosen once and reused while the downstream model or forecasting horizon changes. This paper isolates...

<https://arxiv.org/abs/2609.07533>

3 **CONF****Scoring Without the Engine: Validating a Deterministic, Manipulation-Resistant Content Score for Generative Engines, End to End****ArXiv CS.AI 50%**

arXiv:2609.07559v1 Announce Type: new Abstract: How do you validate a cheap, deterministic proxy for an oracle that is expensive, rate-limited, and non-stationary? We present a protocol built on adversarial falsification gates (negative control, dose response, bounded...

<https://arxiv.org/abs/2609.07559>

4 **CONF****A Tool-Augmented, GPT-4 Chatbot for Real-Time Repository Data Analysis****ArXiv CS.AI 50%**

arXiv:2609.07586v1 Announce Type: new Abstract: Software repositories contain vast amounts of data on code contributions, bug reports, and project activities, yet this information remains challenging for non-technical stakeholders and developers to access due to limited...

<https://arxiv.org/abs/2609.07586>

5 **CONF****Norms at a Price: Why RL-Based Alignment Can Promise Conditional Compliance at Best****ArXiv CS.AI 50%**

arXiv:2609.07627v1 Announce Type: new Abstract: AI agents sometimes act aligned when they infer they are being tested, and differently when not. We argue this is not an anomaly but what current training regimes are structured to select for. Reinforcement-learning-based alignment...

<https://arxiv.org/abs/2609.07627>

6 **CONF****The Emerging AI Paper-Review Arms Race: Adversarial Co-Evolution in Scholarly Publishing****ArXiv CS.AI 50%**

arXiv:2609.07713v1 Announce Type: new Abstract: Generative and agentic AI are reshaping both the production and evaluation of scientific research. These developments are often studied separately, as questions of how AI can produce research and how AI can review it. We argue that...

<https://arxiv.org/abs/2609.07713>

7 CONF

The Profit Alignment Problem: How Profit Mandates Induce Alignment Failures in LLMs

ArXiv CS.AI 50%

arXiv:2609.07731v1 Announce Type: new Abstract: We show that ordinary business language --- "maximize profitability" --- induces profit-oriented ambiguity resolution: LLMs systematically dismiss ambiguous signals of potential safety violations to serve business objectives. In...

<https://arxiv.org/abs/2609.07731>

8 CONF

What Does an LLM-Agent Leaderboard Rank Actually Compare?

ArXiv CS.AI 50%

arXiv:2609.07785v1 Announce Type: new Abstract: An LLM-agent leaderboard invites a familiar inference: an agent ranked above another is the better agent. Public evaluation logs may not support that conclusion when systems differ in task mixture, label source, release detail, or...

<https://arxiv.org/abs/2609.07785>

9 CONF

Understanding the Impact of Model Pruning on Long-Tail Forgetting and Explanation Reliability in Medical Imaging

ArXiv CS.AI 50%

arXiv:2609.07803v1 Announce Type: new Abstract: Model pruning is widely used to compress deep neural networks, reducing memory and computational requirements with minimal impact on aggregate performance. However, its effect on model behavior remains poorly understood,...

<https://arxiv.org/abs/2609.07803>

0 CONF

Do Large Language Models Know What They Don't Know II? A Fully Behavioral, Non-Cognitive Measure of Epistemic Honesty

ArXiv CS.AI 50%

arXiv:2609.07879v1 Announce Type: new Abstract: Large Language Models (LLMs) are frequently confident, eloquent, and well versed. A natural question arises: do they know what they don't know? To answer this question, we borrow the concept of epistemic honesty and develop a novel...

<https://arxiv.org/abs/2609.07879>

1 CONF

When Can LLM Digital Twins Reduce Human Measurement? From Behavioral Fidelity to Statistical Substitutability

ArXiv CS.AI 50%

arXiv:2609.07987v1 Announce Type: new Abstract: LLM-based digital twins promise to reduce repeated human data collection by generating person- specific responses, yet existing evaluations provide little evidence about whether they can reduce human measurement while preserving...

<https://arxiv.org/abs/2609.07987>

2 CONF

From Version Conflicts to Decision Conflicts: Selective Revalidation for Long-Running AI Agents

ArXiv CS.AI 50%

arXiv:2609.08015v1 Announce Type: new Abstract: Long-running AI agents may read state, reason, wait for tools or human approval, and perform an external action much later. The state that justified the action can change in the meantime. For example, after an agent proposes an 80...

<https://arxiv.org/abs/2609.08015>

3 CONF

A Layered Analysis of Disagreement And Answer Quality in Multi-Agent LLM Debate

ArXiv CS.AI 50%

arXiv:2609.08016v1 Announce Type: new Abstract: Multi-agent debate, in which several LLMs exchange arguments before answering, is widely assumed to improve answer quality by surfacing genuine disagreement. That mechanism is rarely checked. We introduce four measurements: (A) the...

<https://arxiv.org/abs/2609.08016>

4 CONF

Automated Design of Inventory Policy with Large Language Models: An Exploratory Study

ArXiv CS.AI 50%

arXiv:2609.08071v1 Announce Type: new Abstract: Firms making inventory decisions have access to operational data, optimization tools, and large language models (LLMs). Typically, data characterize the operating environment, optimization selects parameters within a prespecified...

<https://arxiv.org/abs/2609.08071>

5 CONF

RevalExo: A Functional Daily-Activity Benchmark for Inertial and Visual Locomotion Mode Recognition in Older Adults and Clinical Cohorts

ArXiv CS.AI 50%

arXiv:2609.08090v2 Announce Type: new Abstract: Assistive devices for people with mobility impairments, such as powered exoskeletons, rely on accurate locomotion mode recognition to adapt control strategies and provide appropriate assistance during daily activities. However,...

<https://arxiv.org/abs/2609.08090>

6 CONF

CIVI: A Framework for Diagnosing Search Agent Failures in Civic Information

ArXiv CS.AI 50%

arXiv:2609.08094v1 Announce Type: new Abstract: Large Language Models are increasingly deployed in public-sector settings, where incorrect guidance can cause irreversible harm. We introduce CIVI, the first framework for diagnosing search agent failures in civic information. Its...

<https://arxiv.org/abs/2609.08094>

7 CONF

Artificial Intelligence-Assisted Digital Inventory of Cultural Heritage & Traditional Knowledge: Case for Indonesian Open Digital Library of Culture

ArXiv CS.AI 50%

arXiv:2609.08105v1 Announce Type: new Abstract: The Indonesian Digital Library of Culture (Perpustakaan Digital Budaya Indonesia, PDBI; budaya-indonesia.org) is a participatory platform that has collected tens of thousands of entries on Nusantara cultural heritage through public...

<https://arxiv.org/abs/2609.08105>

8 CONF

OntologyBench: Can Dense Retrieval Satisfy Structured Biomedical Constraints?

ArXiv CS.AI 50%

arXiv:2609.08174v1 Announce Type: new Abstract: We introduce OntologyBench, a tiered biomedical retrieval benchmark comprising 471,854 training and 125,744 evaluation query-document relevance pairs across concept grounding, relational retrieval, and compositional phenotype-based...

<https://arxiv.org/abs/2609.08174>

9 CONF

Less Is Personal: Learning Minimal Sufficient User Profiles for Personalized Language Models

ArXiv CS.AI 50%

arXiv:2609.08180v1 Announce Type: new Abstract: Retrieval-augmented personalization enables large language models to produce more accurate and preference-aligned outputs using relevant records retrieved from user histories. Personalized language models typically prepend a fixed...

<https://arxiv.org/abs/2609.08180>

0 CONF

Does Deeper Reasoning Compromise Alignment? Revealing and Mitigating of Alignment Collapse in Large Reasoning Models

ArXiv CS.AI 50%

arXiv:2609.08186v1 Announce Type: new Abstract: The emergence of Chain-of-Thought (CoT) has established a robust foundation for Large Reasoning Models (LRMs). While deep reasoning is widely believed to enhance safety alignment, the stability of alignment mechanisms under...

<https://arxiv.org/abs/2609.08186>

1 CONF

Bridging the Semantic-Utility Gap in Multimodal RAG via Generator-in-the-Loop Alignment

ArXiv CS.AI 50%

arXiv:2609.08188v1 Announce Type: new Abstract: Vision-language models (VLMs) augmented with retrieval-augmented generation (RAG) benefit from access to external evidence. However, standard retrievers and rerankers optimize for semantic similarity rather than answer utility,...

<https://arxiv.org/abs/2609.08188>

2 CONF

DUA-D2C: Dynamic Uncertainty Aware Method for Overfitting Remediation in Deep Learning

ArXiv CS.AI 50%

arXiv:2411.15876v3 Announce Type: replace-cross Abstract: Overfitting remains a significant challenge in deep learning, often arising from data outliers, noise, and limited training data. To address this, we previously proposed the Divide2Conquer (D2C) method, which partitions...

<https://arxiv.org/abs/2411.15876>

3 CONF

ProcArena: A Multi-Scenario Benchmark for LLMs on Direct and Interactive PL/SQL Development from Natural Language

ArXiv CS.AI 50%

arXiv:2609.06527v1 Announce Type: cross Abstract: Large language models (LLMs) have shown strong potential for translating natural-language (NL) requirements into PL/SQL programs, attracting increasing attention from the database community. However, existing NL-to-PL/SQL efforts...

<https://arxiv.org/abs/2609.06527>

4 CONF

Style Over Substance: Content-Invariant Wrappers Flip LLM Safety-Judge Verdicts

ArXiv CS.AI 50%

arXiv:2609.08236v1 Announce Type: new Abstract: Automatic safety judges -- systems such as Llama Guard or a GPT-4o grading prompt that decide whether a model's reply is harmful -- produce the numbers behind almost every reported jailbreak success rate, defense evaluation, and...

<https://arxiv.org/abs/2609.08236>

5 CONF

AV-SafetyBench: A Safety Benchmark for Text-to-Audio-Video Generation

ArXiv CS.AI 50%

arXiv:2609.06991v1 Announce Type: cross Abstract: Recent text-to-audio-video (T2AV) models jointly generate video, speech, sound effects, and ambience from a single text prompt. This capability poses new challenges for safety evaluation, as unsafe content may be conveyed through...

<https://arxiv.org/abs/2609.06991>

6 CONF

SeRV: Semantic-Aligned Residual Vector Quantization for American Sign Language Generation

ArXiv CS.AI 50%

arXiv:2609.05742v1 Announce Type: cross Abstract: American Sign Language (ASL) generation remains challenging due to limited paired text-ASL motion data and the difficulty of learning motion representations both precise for reconstruction and predictable from linguistic input....

<https://arxiv.org/abs/2609.05742>

7 CONF

FastE: Readout-Triggered Token Compression for LLM Embedding Inference

ArXiv CS.AI 50%

arXiv:2609.08407v1 Announce Type: new Abstract: In this study, we identify depth-dependent prefix redundancy in final-readout LLM embedding models, notably across representative backbones including Qwen3-Embedding and Qwen3-VL-Embedding. We find that removing prefix states is...

<https://arxiv.org/abs/2609.08407>

8 CONF

Feyospace-v1: How the Cyber Mercury Seven Trained Frontier Cyber Models

ArXiv CS.AI 50%

arXiv:2609.08418v1 Announce Type: new Abstract: Training capable cyber agents is often treated primarily as a problem of model scale, yet open-weight post-training is constrained more directly by the cost of executable environments, reliable multi-turn supervision, and access to...

<https://arxiv.org/abs/2609.08418>

9 CONF

A Three-Tier Persona Vector for Controllable User Simulation in Agentic Evaluation

ArXiv CS.AI 50%

arXiv:2609.08592v1 Announce Type: new Abstract: Evaluating tool-augmented LLM agents requires diverse, realistic user inputs yet most evaluation frameworks use flat role descriptions ("you are an angry customer") that produce near-identical conversations regardless of the...

<https://arxiv.org/abs/2609.08592>

00 CONF

Graph-Based Personalized Memory for LLM Agents: Representation, Evolution, Retrieval, and Evaluation

ArXiv CS.AI 50%

arXiv:2609.08599v1 Announce Type: new Abstract: Large Language Model (LLM) agents are evolving from single-session tools toward long-term personal assistants that must adapt to individual users across tasks, contexts, and interactions. This shift makes memory a core requirement...

<https://arxiv.org/abs/2609.08599>

01 CONF

When Can One Obtain Certificates of Optimality Using Positivstellensätze?

ArXiv CS.AI 50%

arXiv:2609.08736v1 Announce Type: new Abstract: We study certificates of positivity and optimality for learning problems whose objectives and constraints need not be polynomial. We isolate an axiomatic core of Fischer's constructive strict and weak Positivstellensatz and...

<https://arxiv.org/abs/2609.08736>

02 CONF

It's All in the Way You Say It: The Role of Information Representation in LLM-Based Glycemic-Event Prediction

ArXiv CS.AI 50%

arXiv:2609.08772v1 Announce Type: new Abstract: Large Language Models (LLMs) are increasingly being investigated for physiological time-series prediction, yet their effectiveness may depend not only on the model itself, but also on how physiological information is represented...

<https://arxiv.org/abs/2609.08772>

03 CONF

API Benchmark Scores Do Not Reliably Transfer to Chatbot Interfaces

ArXiv CS.AI 50%

arXiv:2609.08861v1 Announce Type: new Abstract: Benchmark scores are a central currency in model releases: they inform purchasing decisions, shape public trust, and influence policy. Yet, a key assumption underlying benchmark scores is that the model performance measured through...

<https://arxiv.org/abs/2609.08861>

04 CONF

PlannerForge: LLM Agents for Scenario-Based Testing of Motion Planners in Autonomous Driving

ArXiv CS.AI 50%

arXiv:2609.08965v1 Announce Type: new Abstract: Ensuring the safety of autonomous driving is a critical challenge. Scenario-based testing is a systematic process used to validate Autonomous Driving Systems (ADSs), but it remains a fragmented modular pipeline in which scenario...

<https://arxiv.org/abs/2609.08965>

05 CONF

Good Pretraining, Bad SFT: Checkpoint Quality Across the Training Stack

ArXiv CS.AI 50%

arXiv:2609.08966v1 Announce Type: new Abstract: Language-model checkpoints are commonly selected by pretraining loss or benchmark scores, assuming that the highest-scoring checkpoint will remain the best starting point for subsequent training. We show that this assumption can...

<https://arxiv.org/abs/2609.08966>

06 CONF

Deposon: An Auditable, Conservation-Guaranteed, Game-Theoretically Tested Scattering Layer over LLM Reasoning Paths

ArXiv CS.AI 50%

arXiv:2609.09001v1 Announce Type: new Abstract: Multi-step LLM reasoning lacks a machine-recheckable ledger: discarded reasoning paths leave no auditable record. We propose the Deposon scattering layer, which binds each node of an LLM-generated concept-decomposition graph to a...

<https://arxiv.org/abs/2609.09001>

07 CONF

Answer-Distribution Trajectories: A Stochastic-Dynamics View of LLM Reasoning

ArXiv CS.AI 50%

arXiv:2609.09030v1 Announce Type: new Abstract: Chain-of-thought reasoning provides a structured computation between a model's input and final answer. Yet it is often evaluated through endpoint accuracy, which ignores the path taken to reach that answer. An emerging line of work...

<https://arxiv.org/abs/2609.09030>

08 CONF

Everything in Moderation: Per-Domain Coverage Optima and Alignment-Resistant Domain Gaps in Multi-Domain Mid-Training

ArXiv CS.AI 50%

arXiv:2609.09081v1 Announce Type: new Abstract: Mid-training, the stage between pre-training and alignment, is where a model's per-domain data composition is typically set by data availability rather than principled design. We ask what that decision buys, and whether a later...

<https://arxiv.org/abs/2609.09081>

09 CONF

SAEScientist-Bench: Can AI Agents Conduct Autonomous SAE Interpretability Research?

ArXiv CS.AI 50%

arXiv:2609.09113v1 Announce Type: new Abstract: While research on recursive self-improvement (RSI) has predominantly automated model training pipelines, reliable autonomous development demands a missing pillar: post-hoc monitoring and auditing to understand what models learn and...

<https://arxiv.org/abs/2609.09113>

10 CONF

Co-Evolving Harnesses and Models: On-Policy Correction Helps Weaker Models Catch Up Where Imitation Fails

ArXiv CS.AI 50%

arXiv:2609.09134v1 Announce Type: new Abstract: Agent harnesses (the system prompt, tool set, execution hooks, and context-management scaffolding around a model) are a critical determinant of agentic task success. Automated harness evolution can enable smaller models to perform...

<https://arxiv.org/abs/2609.09134>

11 CONF

Data-driven rational function neural networks: a new method for generating analytical models of rock physics

ArXiv CS.AI 50%

arXiv:2109.08813v1 Announce Type: cross Abstract: Seismic wave velocity of underground rock plays important role in detecting internal structure of the Earth. Rock physics models have long been the focus of predicting wave velocity. However, construction of a theoretical model...

<https://arxiv.org/abs/2109.08813>

12 CONF

Neural Symbolic Regression Using Deep Learning and Sparse Modelling

ArXiv CS.AI 50%

arXiv:2609.01102v1 Announce Type: cross Abstract: Symbolic Regression (SR) seeks to find succinct mathematical expressions that represent the fundamental relationships within data, providing interpretability and scientific understanding that exceeds that of black-box models....

<https://arxiv.org/abs/2609.01102>

13 CONF

TC-Next: Zero-Shot Multimodal Cyclone Forecasting

ArXiv CS.AI 50%

arXiv:2609.02085v1 Announce Type: cross Abstract: We present TropicalCycloneNext (TC-Next), a multimodal deep learning model that forecasts tropical cyclone track and intensity at \$6\$-\$24\$ h leads by leveraging a foundation model's forecast fields of atmospheric kinematic and...

<https://arxiv.org/abs/2609.02085>

14 CONF

When Do Options Help? Policy Necrosis and Redundant Coverage in Option-Critic

ArXiv CS.AI 50%

arXiv:2609.05508v1 Announce Type: cross Abstract: Option-critic learns options: sub-policies together with a learned rule for when each one hands control back. Its headline result is that performance improves as options are added. We explain that result, with theory and...

<https://arxiv.org/abs/2609.05508>

15 CONF

CrossModalQA: A Cross-modal and Multi-hop Benchmark for Multimodal Retrieval-augmented Generation

ArXiv CS.AI 50%

arXiv:2609.05518v1 Announce Type: cross Abstract: Despite the strong capabilities of multimodal large language models (MLLMs), their parametric knowledge remains incomplete and difficult to update, motivating multimodal retrieval-augmented generation (RAG) to ground responses in...

<https://arxiv.org/abs/2609.05518>

16 CONF

DART: A DAG-Based Reputation and Incentive Framework via Blockchain-Enabled Governance for Trustworthy LLM Multi-Agent Collaboration

ArXiv CS.AI 50%

arXiv:2609.05529v1 Announce Type: cross Abstract: Large language model (LLM)-based multi-agent systems (MAS) predominantly rely on centralized orchestration and lack formal verification mechanisms for agent reliability, participation, and system-level behavioral alignment. These...

<https://arxiv.org/abs/2609.05529>

17 CONF

An Agent Model Abstraction for Human-AI Teaming Cognitive Coupling

ArXiv CS.AI 50%

arXiv:2609.05552v1 Announce Type: cross Abstract: Industrial environments increasingly rely on collaboration between humans and AI-enabled agents. Effective teamwork requires aligning how agents perceive situations, plan actions to pursue goals, and adapt to changing conditions,...

<https://arxiv.org/abs/2609.05552>

18 CONF

ViT3Flow: A Test-Time Training Transformer MeanFlow for Postoperative Radiograph Synthesis in Scoliosis

ArXiv CS.AI 50%

arXiv:2609.05579v1 Announce Type: cross Abstract: Predicting postoperative spinal morphology from preoperative radiographs could provide valuable support for scoliosis surgical planning, but remains challenging because surgical correction induces large spatial changes while...

<https://arxiv.org/abs/2609.05579>

19 CONF

Who Maintains Agent Skills? A Longitudinal Study of Human-Governed, AI-Assisted Skill Maintenance

ArXiv CS.AI 50%

arXiv:2609.05677v1 Announce Type: cross Abstract: Lifelong LLM agents increasingly rely on external skill artifacts as one element for preserving and reusing capabilities over time. These skills (usually portable Markdown files such as SKILL.md) describe when and how to apply a...

<https://arxiv.org/abs/2609.05677>

20 CONF

Analysis of Respiratory Sinus Arrhythmia with Neural Networks

ArXiv CS.AI 50%

arXiv:2609.05698v1 Announce Type: cross Abstract: The paper introduces a neural network-based approach for analyzing ECG signals to estimate respiratory rate by leveraging the phenomenon of Respiratory Sinus Arrhythmia (RSA). Our method employs a deep learning model trained to...

<https://arxiv.org/abs/2609.05698>

21 CONF

XAI-SDN: An Explainable Entropy-Guided Machine Learning Framework for Real-Time DDoS Detection in Software Defined Networks

ArXiv CS.AI 50%

arXiv:2609.05701v1 Announce Type: cross Abstract: One of the biggest risks faced by Software Defined Networks (SDN) is the Distributed Denial of Service (DDoS) attack in which a compromised controller can make an entire network unusable. To address these challenges, we suggest...

<https://arxiv.org/abs/2609.05701>

22 CONF

GeoContext: One Context Ladder, Two Failure Modes in Vision-Language Geolocation: Flat Reliance on User-Provided Location Context and False Confirmation of Location Claims

ArXiv CS.AI 50%

arXiv:2609.05761v1 Announce Type: cross Abstract: Visual geolocation benchmarks typically ask a model where an image was captured without accounting for the location context that users often provide. We introduce GeoContext, a resource supporting two complementary tasks:...

<https://arxiv.org/abs/2609.05761>

23 CONF

Interface-Aware KV Cache Quantization for Dense On-Chip NVM in Long-Context LLM Decoding

ArXiv CS.AI 50%

arXiv:2609.05764v1 Announce Type: cross Abstract: The key-value (KV) cache is the dominant memory bottleneck in long-context large language model (LLM) decoding: every step reads it entirely, so decoding is memory-bandwidth bound. Holding a quantized KV cache in dense on-chip...

<https://arxiv.org/abs/2609.05764>

24 CONF

Data Scout: Targeted Web Crawling for Domain-Specific Pretraining Corpora

ArXiv CS.AI 50%

arXiv:2609.05766v1 Announce Type: cross Abstract: The dominant approach to building domain-specific pretraining corpora is to filter large web archives such as CommonCrawl. This works well for popular domains but breaks down for specialized ones, where relevant content is sparse...

<https://arxiv.org/abs/2609.05766>

25 **CONF****Closed-Loop Evaluation of Bird's-Eye-View Maps from Cross-View Transformers as Inputs to Behavior-Cloning Policies****ArXiv CS.AI** **50%**

arXiv:2609.05783v1 Announce Type: cross Abstract: In autonomous driving, Bird's-Eye View (BEV) representations provide a structured, top-down abstraction of the vehicle's surroundings and have become a key input modality for Behavioral Cloning (BC) policies. While ground-truth...

<https://arxiv.org/abs/2609.05783>

26 **CONF****Do Quantum AIs Dream in Paths? Path-Integral Slow Thinking through Grover Interference****ArXiv CS.AI** **50%**

arXiv:2609.05842v1 Announce Type: cross Abstract: Reinforcement learning with verifiable rewards enables large language models to think slowly, but the same training can induce policy collapse: probability concentrates onto a few successful trajectories and exploratory diversity...

<https://arxiv.org/abs/2609.05842>

27 **CONF****SAFEGuard: Detect Optimization-Based Jailbreak Attacks Through Harmful Semantic Analysis and Fluency Measurement****ArXiv CS.AI** **50%**

arXiv:2609.05850v1 Announce Type: cross Abstract: Despite the significant efforts devoted to aligning large language models (LLMs) with human values and ensuring safe deployment, recent work has revealed that LLMs remain vulnerable to adversarial jailbreak attacks that can...

<https://arxiv.org/abs/2609.05850>

28 **CONF****Temporal Heterogeneous Graph Transformer for Credit Card Fraud Detection****ArXiv CS.AI** **50%**

arXiv:2609.07100v1 Announce Type: cross Abstract: Credit card fraud detection typically relies on tabular features, while repeated attributes can also provide useful relational signals. This paper proposes THGT-FD, a Temporal Heterogeneous Graph Transformer for Fraud Detection...

<https://arxiv.org/abs/2609.07100>

29 **CONF****What if LLMs Ate Their Words: Causal History Effects in Multi-Turn Interaction****ArXiv CS.AI** **50%**

arXiv:2609.05882v1 Announce Type: cross Abstract: Multi-turn interaction creates a feedback process in which an LLM's previous responses become context for later behavior. Prior work shows substantial multi-turn degradation and that assistant-generated history can affect later...

<https://arxiv.org/abs/2609.05882>

30 **CONF****From Review to Authorization: Key-Isolated Threshold Signing for LLM Agents****ArXiv CS.AI** **50%**

arXiv:2609.05901v1 Announce Type: cross Abstract: Autonomous LLM agents can turn untrusted content into effectful actions such as payments and permission changes. If the same process interprets this content and controls a reusable signing credential, prompt injection can cross...

<https://arxiv.org/abs/2609.05901>

31 CONF

Structurally Close, Temporally Distant: Measuring Security Exposure in Long-Horizon LLM Agents

ArXiv CS.AI 50%

arXiv:2609.05911v1 Announce Type: cross Abstract: Long-horizon LLM agents interact with untrusted content, persistent memory, external state, and sensitive tools. Existing analyses often characterize attacks by the number of execution steps between malicious input and a...

<https://arxiv.org/abs/2609.05911>

32 CONF

AdaExplore: Failure-Driven Adaptation and Diversity-Preserving Search for Efficient Kernel Generation

ArXiv CS.AI 50%

arXiv:2604.16625v2 Announce Type: replace-cross Abstract: Recent large language model (LLM) agents have shown promise in using execution feedback for test-time adaptation. However, robust self-improvement remains far from solved: most approaches still treat each problem instance...

<https://arxiv.org/abs/2604.16625>

33 CONF

Tri-PvP: Exposing Modality Bias in Omni-Modal Large Language Models through Perceptual-Propositional Evidence Conflicts

ArXiv CS.AI 50%

arXiv:2609.06011v1 Announce Type: cross Abstract: Omni-modal large language models (OLLMs) jointly process vision, audio, and text, yet their modality bias under cross-modal conflict remains underexplored. Existing benchmarks conflate two distinct forms of evidence within a...

<https://arxiv.org/abs/2609.06011>

34 CONF

The Role of Gradient Modification in Heavy-Tailed Nonconvex Stochastic Min-Max Optimization

ArXiv CS.AI 50%

arXiv:2609.06064v1 Announce Type: cross Abstract: Stochastic min-max optimization has attracted increasing attention due to its applications in modern machine learning, while existing theoretical studies mainly rely on the bounded variance assumption for stochastic gradients....

<https://arxiv.org/abs/2609.06064>

35 CONF

FedSubMuon: Communication-Efficient Federated LLM Fine-Tuning via Structured Subspace Muon

ArXiv CS.AI 50%

arXiv:2609.06073v1 Announce Type: cross Abstract: Federated fine-tuning adapts large language models (LLMs) to decentralized client data, but its scalability in cross-device training is often limited by the high communication cost. Muon is an optimizer that improves optimization...

<https://arxiv.org/abs/2609.06073>

36 CONF

Explainable Deep Learning for Price-Trade Dynamics: From Black-Box Forecasts to Effective Parametric Models

ArXiv CS.AI 50%

arXiv:2609.06085v1 Announce Type: cross Abstract: Understanding the joint dynamics of prices and trades is central to market microstructure, where returns and order flow interact through nonlinear and state-dependent mechanisms. Linear models are interpretable but may miss these...

<https://arxiv.org/abs/2609.06085>

37 CONF

Flawed but Memorable: Student Critical Reception of Interest-Personalized GenAI Analogies in Computing Education

ArXiv CS.AI 50%

arXiv:2609.06095v1 Announce Type: cross Abstract: Motivation: Undergraduate computing students increasingly turn to generative AI (GenAI) tools to understand abstract concepts through analogies. Analogies compare an unfamiliar concept to something familiar, but judging whether...

<https://arxiv.org/abs/2609.06095>

38 CONF

PAGR: Proof-Carrying Algebraic-Geometric Retrieval: A Quiver-, Provenance-, and Sheaf-Theoretic Framework for Grounded LLM Retrieval

ArXiv CS.AI 50%

arXiv:2609.06127v1 Announce Type: cross Abstract: Retrieval-augmented generation is usually formulated as a statistical information-retrieval problem. Graph-based variants add relational structure, but the mathematical status of that structure is often left underspecified. Three...

<https://arxiv.org/abs/2609.06127>

39 CONF

What the Window Does Not Contain: Auditing Provenance in a Document-Grounded Instability Benchmark

ArXiv CS.AI 50%

arXiv:2609.06147v1 Announce Type: cross Abstract: Ask a language model the same question about the same document twenty times, and it sometimes returns two different answers. We built Probity, a benchmark of 60 tasks and 470 items from real venture-financing filings, to measure...

<https://arxiv.org/abs/2609.06147>

40 CONF

ExpertLens: Visualizing Embedding Spaces for Post-Hoc Explainability in MoE Enhanced Retrievers

ArXiv CS.AI 50%

arXiv:2609.06155v1 Announce Type: cross Abstract: Neural models, including dense retrievers, have been widely adopted in Information Retrieval (IR), often delivering state-of-the-art performance. Despite their effectiveness, these models operate as black boxes, limiting the...

<https://arxiv.org/abs/2609.06155>

41 CONF

All for 1-Bit: Towards Genuine 1-Bit Post-Training Quantization for LLMs

ArXiv CS.AI 50%

arXiv:2609.06161v1 Announce Type: cross Abstract: Large language models (LLMs) have achieved remarkable progress, yet their massive storage and memory-bandwidth demands still hinder efficient deployment. Weight binarization is a promising solution, but existing...

<https://arxiv.org/abs/2609.06161>

42 CONF

Multiple Myeloma Lesion Segmentation on Whole-Body Diffusion-Weighted Imaging via Efficient Anatomical Anticipation and Multimodal Confirmation

ArXiv CS.AI 50%

arXiv:2609.06165v1 Announce Type: cross Abstract: Whole-body diffusion-weighted imaging (WB-DWI) is widely used for multiple myeloma (MM) assessment, yet automated lesion segmentation remains challenging due to limited anatomical delineation and the low specificity of marrow...

<https://arxiv.org/abs/2609.06165>

43 CONF

Scratchy: Visual-Scratchpad Multimodal Reasoning for Cryptographic Proof Generation in EasyCrypt

ArXiv CS.AI 50%

arXiv:2609.06226v1 Announce Type: cross Abstract: Large language models (LLMs) have recently made substantial progress in formal proof generation, yet presenting distinctive challenges in cryptographic area. Computational security arguments posit that a valid proof must...

<https://arxiv.org/abs/2609.06226>

44 CONF

It is Not Yet Another Tool: Creating and Deploying an Agentic AI Companion in a Security Operations Center

ArXiv CS.AI 50%

arXiv:2609.06250v1 Announce Type: cross Abstract: Security Operations Centers (SOCs) process large amounts of tickets, most of which are low-interest events not worthy of further investigation. The repetitive nature of this task and similarity of the vast amounts of tickets make...

<https://arxiv.org/abs/2609.06250>

45 CONF

Steering Geometry: Validating Human Value Geometry in LLM Steering Space

ArXiv CS.AI 50%

arXiv:2609.06289v1 Announce Type: cross Abstract: As large language models (LLMs) are increasingly deployed in alignment-sensitive contexts, activation steering has emerged as a lightweight, inference-time alternative to fine-tuning methods (e.g., RLHF, DPO) for behavioral...

<https://arxiv.org/abs/2609.06289>

46 CONF

Robust Conformal Consensus: Multi-Agent LLM-as-a-Judge Interval Evaluation with Conformal Prediction

ArXiv CS.AI 50%

arXiv:2609.06367v1 Announce Type: cross Abstract: LLM-as-a-Judge has emerged as a promising paradigm for evaluating natural language generation. However, the uncertainty associated with such evaluations remains largely unexplored, which limits their reliability in real-world...

<https://arxiv.org/abs/2609.06367>

47 CONF

Parameterized and Streaming Algorithms for Euclidean Fair k -Center Clustering

ArXiv CS.AI 50%

arXiv:2609.06384v1 Announce Type: cross Abstract: Motivated by the growing importance of fairness in machine learning, fair k -center clustering has attracted considerable research attention as a fundamental problem. In this problem, a dataset is partitioned into m disjoint...

<https://arxiv.org/abs/2609.06384>

48 CONF

VICT: Verifier-Instrumented Credit Tracing for Long-Horizon LLM Agent Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2608.28128v2 Announce Type: replace-cross Abstract: Fine-grained credit assignment is a central challenge in reinforcement learning for long horizon LLM agents. Standard objectives often train from programmatically verifiable terminal rewards by broadcasting each sparse...

<https://arxiv.org/abs/2608.28128>

49 CONF

One MLLM, One Call: Efficient Zero-Shot Vision-and-Language Navigation via Spatial-Aware Waypoints

ArXiv CS.AI 50%

arXiv:2609.06476v1 Announce Type: cross Abstract: Vision-and-Language Navigation in Continuous Environments (VLN-CE) requires an embodied agent to navigate unseen environments by following natural language instructions. Current zero-shot VLN-CE methods either rely on pre-trained...

<https://arxiv.org/abs/2609.06476>

50 CONF

Second-Order Smooth Planning with Optimal-Transport Bellman Smoothing

ArXiv CS.AI 50%

arXiv:2609.06484v1 Announce Type: cross Abstract: Planning with a generative model aims to estimate the value of a state using as few simulator calls as possible. SmoothCruiser achieves problem-independent complexity $O(\epsilon^{-4})$ by exploiting the smoothness...

<https://arxiv.org/abs/2609.06484>

51 CONF

OracleZoom: On-Policy Self-Distillation Inspired Reference-Constrained Recursive Image Super Resolution

ArXiv CS.AI 50%

arXiv:2609.06490v1 Announce Type: cross Abstract: Recursive Super-Resolution (SR) extends fixed-scale SR to extreme magnification by repeatedly feeding predictions back into the same model, analogous to zooming an image repeatedly. However, ground truth availability at every...

<https://arxiv.org/abs/2609.06490>

52 CONF

Layer-Wise Gate-Controlled Prompt Truncation in a Multimodal Chest X-Ray Classifier

ArXiv CS.AI 50%

arXiv:2609.06590v1 Announce Type: cross Abstract: Mixture of Prompt Experts (MoPE) adapts multimodal transformers through input-dependent prompt composition, while retaining a fixed prompt length. We investigate a layer-wise gating extension in a binary chest X-ray...

<https://arxiv.org/abs/2609.06590>

53 CONF

MemCorr-DP: Counterfactual Correspondence Conditioning for a Diffusion Policy Guided by a Reference

ArXiv CS.AI 50%

arXiv:2609.06615v1 Announce Type: cross Abstract: Behavior-cloned visuomotor policies can remain accurate near their training distribution yet fail when object position and camera viewpoint change together. A successful reference trajectory contains the geometry needed to...

<https://arxiv.org/abs/2609.06615>

54 CONF

Mind the Gap: Exposing LLM Translation Blind Spots Using the AlphaMWE Multilingual Parallel Corpus

ArXiv CS.AI 50%

arXiv:2609.06634v1 Announce Type: cross Abstract: LLMs' performance on machine translation (MT) tasks is often dependent on the data availability in the specific domains and language pairs that they are trained upon. To examine if Multiword Expressions (MWEs) still set a...

<https://arxiv.org/abs/2609.06634>

55 CONF

Inducing Emergent Misalignment from Reward Hacks with Iterative DPO

ArXiv CS.AI 50%

arXiv:2609.06649v1 Announce Type: cross Abstract: Reward hacking during reinforcement learning from verifiable rewards (RLVR) can induce reward seeking and broad misalignment in language models. Studying this misgeneralization is important for developing better threat models and...

<https://arxiv.org/abs/2609.06649>

56 CONF

KTO: Model Alignment as Prospect Theoretic Optimization

ArXiv CS.AI 50%

arXiv:2402.01306v5 Announce Type: replace-cross Abstract: Kahneman & Tversky's $\textit{prospect theory}$ tells us that humans perceive random variables in a biased but well-defined manner (1992); for example, humans are famously loss-averse. We show that objectives for aligning...

<https://arxiv.org/abs/2402.01306>

57 CONF

Assessing Covariate-Informed Grid Load Forecasting with a Time-Series Foundation Model

ArXiv CS.AI 50%

arXiv:2609.06656v1 Announce Type: cross Abstract: Modern power systems are growing increasingly complex as they integrate diverse generation sources to meet rising demand, making accurate load forecasting challenging. Recent advances in time-series foundation models (TSFMs)...

<https://arxiv.org/abs/2609.06656>

58 CONF

Detokenization Leaks: Reconstructing Local LLM Outputs From Cache Traces

ArXiv CS.AI 50%

arXiv:2609.06674v1 Announce Type: cross Abstract: We present a new attack that reconstructs the text generated by locally hosted LLMs by observing CPU cache activity during detokenization. Unlike prior attacks that rely on deployment-specific assumptions, such as shared data...

<https://arxiv.org/abs/2609.06674>

59 CONF

Counterfactual Tests for Measuring Chain-of-Thought Faithfulness in Visual Language Models

ArXiv CS.AI 50%

arXiv:2609.06704v1 Announce Type: cross Abstract: Chain-of-thought (CoT) may often look plausible, yet it may not faithfully reflect the model's decision-making process. While methods for measuring the faithfulness of CoTs for textual inputs have been increasingly introduced,...

<https://arxiv.org/abs/2609.06704>

60 CONF

A Trustworthy Watermarking Framework for LLM-Generated Food Safety Content

ArXiv CS.AI 50%

arXiv:2609.06708v1 Announce Type: cross Abstract: Large language models are transforming many industries with their text generation abilities. However, their outputs can be easily tampered with, creating serious risks in critical areas such as food safety reporting. To protect...

<https://arxiv.org/abs/2609.06708>

61 CONF

Ordinary, Reasonable Chatbots: Do AI Models Track Human Legal Judgments?

ArXiv CS.AI 50%

arXiv:2609.06769v1 Announce Type: cross Abstract: As people increasingly rely on artificial intelligence (AI) for guidance in their own lives, scholars, lawyers, and even judges have begun to consider the role of AI in legal decision-making. As "silicon sampling" -- the use of...

<https://arxiv.org/abs/2609.06769>

62 CONF

Hardware Trojan Threats to Multi-Chiplet Photonic Neural Network Accelerators

ArXiv CS.AI 50%

arXiv:2609.06796v1 Announce Type: cross Abstract: Multi-chiplet photonic neural network accelerators (MCPNAs) combine photonic computation, photonic communication, and heterogeneous chiplet integration to enable scalable and energy-efficient AI acceleration. However, their...

<https://arxiv.org/abs/2609.06796>

63 CONF

From Synthetic Priors to Model Behavior: Structural Coverage in Tabular Foundation Models

ArXiv CS.AI 50%

arXiv:2609.06912v1 Announce Type: cross Abstract: Tabular foundation models (TFMs) are commonly pretrained on large collections of procedurally generated synthetic tasks, yet it remains unclear how well these synthetic pretraining priors support the downstream tasks on which the...

<https://arxiv.org/abs/2609.06912>

64 CONF

Typed Federated Artifacts for the Agentic Web: Sharing Tool-Routing Knowledge Across Frozen, Heterogeneous LLM Agents

ArXiv CS.AI 50%

arXiv:2609.06815v1 Announce Type: cross Abstract: An open, networked web will allow agents to run frozen models from multiple vendors, keep their history private, and teach each other which tool to call and when. Flat text (prompts, example pools) makes it difficult for the...

<https://arxiv.org/abs/2609.06815>

65 CONF

XYBench: Can LLMs Respond Pragmatically to Queries with Misconceptions?

ArXiv CS.AI 50%

arXiv:2609.06842v1 Announce Type: cross Abstract: When non-expert users ask LLMs for assistance, their queries can often have misconceptions (e.g., "How do I parse XML with regex?"). In such cases, often referred to as the XY-problem, LLMs must identify the misconception ("regex...

<https://arxiv.org/abs/2609.06842>

66 CONF

Characterizing Contention-Induced Reliability Collapse in KV-Cache Timing Side Channels for Multi-Tenant LLM Serving

ArXiv CS.AI 50%

arXiv:2609.06853v1 Announce Type: cross Abstract: Shared key-value (KV) cache reuse improves large language model (LLM) serving, but it can also create a timing side channel that reveals whether a prefix is already cached. Previous work shows that such attacks are possible, but...

<https://arxiv.org/abs/2609.06853>

67 CONF

Human-agent discovery of reconfigurable in-plane ferroelectric superdomain control

ArXiv CS.AI 50%

arXiv:2609.06887v1 Announce Type: cross Abstract: Automated experimentation is most effective when the observables, available actions, and objective are defined before the experiment starts, as is the case for Bayesian optimization. However, in many exploratory experiments, the...

<https://arxiv.org/abs/2609.06887>

68 CONF

The Geometry of Refusal: Why Post-Hoc Safety Is Fragile and Pretraining-Time Safety Persists

ArXiv CS.AI 50%

arXiv:2609.06934v1 Announce Type: cross Abstract: Post-hoc safety training (RLHF, DPO) is the dominant way to align large language models, yet jailbreaks (Zou et al., 2023b), fine-tuning attacks (Qi et al., 2024), and activation-space probes (Arditi et al., 2024) keep recovering...

<https://arxiv.org/abs/2609.06934>

69 CONF

PCSDiff: Diffusion-Based Bias Correction and Super Resolution Toward Practical Operational Medium-Term Precipitation Forecast

ArXiv CS.AI 50%

arXiv:2609.06942v1 Announce Type: cross Abstract: Medium-range precipitation forecasts are impaired by persistent systematic biases, lead-time-dependent error accumulation, and coarse spatial resolution, restricting their reliability for flood-drought risk assessment. Existing...

<https://arxiv.org/abs/2609.06942>

70 CONF

Re-calibrated Contrastive Loss for Transformation-Aware Prompt Conditioning in Vision-Language Models

ArXiv CS.AI 50%

arXiv:2609.06967v1 Announce Type: cross Abstract: Ensuring effective transfer learning for vision-language models without compromising their generalization performance is crucial. However, many existing methods overlook data characteristics and simply reuse the training...

<https://arxiv.org/abs/2609.06967>

71 CONF

Train Overcomplete, Deploy Compact: Scaling Recovery Capacity for Structured LLM Pruning

ArXiv CS.AI 50%

arXiv:2609.06974v1 Announce Type: cross Abstract: Large language models achieve strong performance across diverse tasks, but deployment remains costly because of memory, latency, and energy demands. Structured pruning reduces these costs by removing architectural components, yet...

<https://arxiv.org/abs/2609.06974>

72 CONF

Diffusion Model in Latent Space for Medical Image Segmentation Task

ArXiv CS.AI 50%

arXiv:2512.01292v4 Announce Type: replace-cross Abstract: Medical image segmentation is crucial for clinical diagnosis and treatment planning. Traditional methods typically produce a single segmentation mask, failing to capture inherent uncertainty. Recent generative models...

<https://arxiv.org/abs/2512.01292>

73 CONF

Aha-Flow Distillation: Flow Markers Matter in LLM Reasoning

ArXiv CS.AI 50%

arXiv:2609.07036v1 Announce Type: cross Abstract: We identify the Flow Moment, a reasoning pattern characterized by sustained, process-confirming verbalizations such as I'm doing, in contrast to the revision- and backtracking-oriented Aha Moment. We refer to their corresponding...

<https://arxiv.org/abs/2609.07036>

74 CONF

From LLM-Generated Specifications to Learned Quadruped Locomotion

ArXiv CS.AI 50%

arXiv:2609.07111v1 Announce Type: cross Abstract: Quadruped robot locomotion policies are often trained using reinforcement learning, which in turn relies heavily on hand-crafted reward functions. Designing reward functions requires substantial manual engineering, and it is...

<https://arxiv.org/abs/2609.07111>

75 CONF

AgentLeak: Cloning Stronger LLM Agent Capabilities onto Weaker Agents Beyond Skill Stealing

ArXiv CS.AI 50%

arXiv:2609.07131v1 Announce Type: cross Abstract: Large language model (LLM) agents increasingly achieve long-horizon tasks by combining foundation models with explicit skills and implicit procedural knowledge acquired through execution. The resulting task-solving capabilities...

<https://arxiv.org/abs/2609.07131>

76 CONF

Flow3D-OPD: Multi-Teacher On-Policy Distillation for 3D Geometry Generation with Flow-Matching Diffusion Transformer

ArXiv CS.AI 50%

arXiv:2609.07137v1 Announce Type: cross Abstract: Recent image-to-3D generation models built on flow-matching diffusion Transformers (DiT) can produce high-fidelity meshes, yet their post-training strategy remains largely unexplored. There exist several critical bottlenecks in...

<https://arxiv.org/abs/2609.07137>

77 CONF

FreqBLiMP: Frequency-Controlled Minimal Pairs Reveal Robustness and Fragility of LLMs Under Lexical Rarity

ArXiv CS.AI 50%

arXiv:2609.07153v1 Announce Type: cross Abstract: Minimal-pair benchmarks such as BLiMP evaluate linguistic knowledge by testing whether language models (LMs) prefer acceptable sentences over minimally different unacceptable ones. However, these benchmarks largely ignore lexical...

<https://arxiv.org/abs/2609.07153>

78 CONF

Deep Learning for Biopsy-Free Subtyping of Basal Cell Carcinoma from Dermatoscopic Images

ArXiv CS.AI 50%

arXiv:2609.07180v1 Announce Type: cross Abstract: Basal Cell Carcinoma (BCC) is the most common type of skin cancer, accounting for nearly 80% of skin cancer diagnoses. Its optimal clinical management is guided by the distinct histopathologic subtype, with aggressive variants...

<https://arxiv.org/abs/2609.07180>

79 CONF

Recompilation Is Not Enough: Test-Guided Decompiled-C Repair

ArXiv CS.AI 50%

arXiv:2609.07201v1 Announce Type: cross Abstract: Decompiled C often becomes recompilable only after repair, but recompilation alone does not establish test-observed behavior. A recompiled command-line binary can still parse options incorrectly, print different bytes, or return...

<https://arxiv.org/abs/2609.07201>

80 CONF

Mathematical Programming in Machine Learning and Artificial Intelligence: A Unified Taxonomy of Models and Applications

ArXiv CS.AI 50%

arXiv:2609.07254v1 Announce Type: cross Abstract: Mathematical programming provides a common language for many decisions embedded in modern machine-learning (ML) and artificial-intelligence (AI) systems: selecting retrieval context, routing tokens, allocating inference compute,...

<https://arxiv.org/abs/2609.07254>

81 CONF

LANTERN: Language Model Assessment on Noisy and Transformed Tasks for Understanding Error and Robustness Nuances

ArXiv CS.AI 50%

arXiv:2609.07309v1 Announce Type: cross Abstract: Robustness evaluation of large language models (LLMs) remains a critical challenge, particularly in assessing their sensitivity to perturbations in input data. In this work, we systematically evaluate LLM robustness across...

<https://arxiv.org/abs/2609.07309>

82 CONF

Monadic Second-Order Logic in HOL: Deep and Shallow with Automated Faithfulness (Extended Preprint)

ArXiv CS.AI 50%

arXiv:2609.07345v1 Announce Type: cross Abstract: In Isabelle/HOL, we apply the deep-and-shallow embedding methodology of our prior work to monadic second-order logic (MSO). Three embeddings are developed side by side: a deep embedding (an inductive datatype with an explicit...

<https://arxiv.org/abs/2609.07345>

83 CONF

BlueprintAgent: Constraint-Triggered Targeted Revisits for Simulation-Ready Generation from Scanned Structural Blueprints

ArXiv CS.AI 50%

arXiv:2609.07362v1 Announce Type: cross Abstract: Converting in-service reinforced-concrete (RC) building blueprints into simulation-ready models---structured frame representations that support deterministic FEM export and qualified-engineer review---underpins safety assessment...

<https://arxiv.org/abs/2609.07362>

84 CONF

Beyond Single-Negative Preference: Multi-Negative DPO for LLM-Centric Historical Entity Linking

ArXiv CS.AI 50%

arXiv:2609.07379v1 Announce Type: cross Abstract: Large language models (LLMs) have recently shown promise for historical entity linking, but preference optimization for this task is often formulated with only one negative candidate per training instance. This discards...

<https://arxiv.org/abs/2609.07379>

85 CONF

Federated Binary Gating with Server-Side Vision-Language Inference for Surveillance Anomaly Classification

ArXiv CS.AI 50%

arXiv:2609.07403v1 Announce Type: cross Abstract: Privacy-sensitive surveillance systems could benefit from large vision-language models (VLMs), but such models typically require centralized access to raw video. In federated learning settings, this challenge is amplified by...

<https://arxiv.org/abs/2609.07403>

86 CONF

WildRelight: A Real-World Benchmark and Physics-Guided Adaptation for Single-Image Relighting

ArXiv CS.AI 50%

arXiv:2605.11696v2 Announce Type: replace-cross Abstract: Recent single-image relighting methods, powered by advanced generative models, have achieved impressive photorealism on synthetic benchmarks. However, their effectiveness in the complex visual landscape of the real world...

<https://arxiv.org/abs/2605.11696>

87 CONF

Generation of Vectorized Maps Beyond Vehicle View

ArXiv CS.AI 50%

arXiv:2609.07511v1 Announce Type: cross Abstract: Autonomous driving relies on High Definition (HD) maps for safe navigation. Traditional HD maps construction is costly in hardware, data and human resources, which together with its update limitations hinders scalability. Recent...

<https://arxiv.org/abs/2609.07511>

88 CONF

Zero-Shot Sim-to-Real Contact-Rich Assembly via Proprioception-Anchored Cross-Modal Pretraining

ArXiv CS.AI 50%

arXiv:2609.07534v1 Announce Type: cross Abstract: Contact-rich assembly remains challenging because it requires submillimeter spatial accuracy and reliable interpretation of forces during sustained contact. Although simulation-based reinforcement learning offers a scalable...

<https://arxiv.org/abs/2609.07534>

89 CONF

We're Cooked! - Probing LLM Political Alignment Via Conflict-Framed Recipe Translation

ArXiv CS.AI 50%

arXiv:2609.07568v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly deployed for translation tasks, yet their implicit political positioning in such contexts remains understudied. We ask whether a single politically charged framing term, such as...

<https://arxiv.org/abs/2609.07568>

90 CONF

Large-Scale User Behavior Analysis in Multimodal AI-Assisted Manual Task Execution

ArXiv CS.AI 50%

arXiv:2609.07594v1 Announce Type: cross Abstract: Conversational Task Assistants (CTAs) are multimodal dialogue systems that support users in complex real-world tasks such as cooking and DIY through voice, text, image, and video interactions. Prior user studies have focused on...

<https://arxiv.org/abs/2609.07594>

91 CONF

Online Surrogate Repair: Decoupling High-Fidelity Feedback from Search Length in Closed-Loop Discovery

ArXiv CS.AI 50%

arXiv:2609.07655v1 Announce Type: cross Abstract: Closed-loop AI scientists can generate candidate designs at low marginal computational cost, whereas reliable feedback may require wet-lab synthesis, characterization, or high-fidelity computation. Addressing this imbalance...

<https://arxiv.org/abs/2609.07655>

92 CONF

Thermodynamic Cyclic Processes with Markov Samplers in Bayesian Inference

ArXiv CS.AI 50%

arXiv:2609.07660v1 Announce Type: cross Abstract: The concept of Markov chain Monte Carlo (MCMC) cycles, an analogy to cyclic processes in heat engines, is presented in order to examine Bayesian inference problems. In this effort, we develop adaptive ensemble schedulers that...

<https://arxiv.org/abs/2609.07660>

93 CONF

How AI Models Manage Epistemic Authority: A Taxonomy and Comparative Analysis of Responses to User Disagreement

ArXiv CS.AI 50%

arXiv:2609.07662v1 Announce Type: cross Abstract: Large language models are increasingly used as sources of advice and information, including in high-stakes settings, yet little is known about how they respond to user disagreement. We study how a model manages its epistemic...

<https://arxiv.org/abs/2609.07662>

94 CONF

Accuracy is Not Enough: A Divergence-Based Approach to Evaluate Fidelity Loss in Quantized LLMs

ArXiv CS.AI 50%

arXiv:2609.07664v2 Announce Type: cross Abstract: Deployment of Large Language Models (LLMs) on memory-constrained edge devices relies heavily on aggressive post-training quantization. However, evaluating these models is largely based on zero-shot task accuracy, which depends...

<https://arxiv.org/abs/2609.07664>

95 CONF

From Citations to Contributions: LLM-Assisted Credit Scoring of Research Articles

ArXiv CS.AI 50%

arXiv:2609.07673v1 Announce Type: cross Abstract: Citation-based measures of scientific influence typically treat citations as uniform signals, ignoring the different roles that cited works play in a paper's contribution. We introduce contribution-based credit scoring for...

<https://arxiv.org/abs/2609.07673>

96 CONF

Do AI Coding Assistants Check Before They Install? A Pre-Registered Demand-Side Audit of Trust Signals in the Research Software Supply Chain

ArXiv CS.AI 50%

arXiv:2609.07754v1 Announce Type: cross Abstract: AI coding assistants now select, install, and configure software, and attackers have exploited that position through invented package names, compromised maintainer accounts, and manipulated repository text. In response, the...

<https://arxiv.org/abs/2609.07754>

97 CONF

Bag of Tricks or Bag of Myths? Reducing Modeling Complexity with Task Knowledge in Explainable Suicide Risk Assessment

ArXiv CS.AI 50%

arXiv:2609.07766v1 Announce Type: cross Abstract: Assessing suicide risk from social media text is a small-data, high-stakes setting requiring not only severity prediction but also supporting evidence and clinically relevant risk and protective factors. Yet common NLP...

<https://arxiv.org/abs/2609.07766>

98 CONF

You Can't Prefer Emotions You Don't Sample: Intensity Undershoot in DPO-Tuned LLMs

ArXiv CS.AI 50%

arXiv:2609.07808v1 Announce Type: cross Abstract: Ask a language model to respond "very excitedly," and its output is typically only mildly more energetic. We quantify this effect. We condition an instruction-tuned LLM on a continuous Valence-Arousal (VA) target, where valence...

<https://arxiv.org/abs/2609.07808>

99 CONF

A*-Thought-V2: Efficient Latent Reasoning via Geometric Dynamics of LLM

ArXiv CS.AI 50%

arXiv:2609.07821v1 Announce Type: cross Abstract: Chain-of-Thought (CoT) improves the reasoning ability of Large Language Models (LLMs) but incurs substantial computation and context costs. Existing methods either lose intermediate information through hard pruning or lack a...

<https://arxiv.org/abs/2609.07821>

00 CONF

AI for AI: Optimizing Additional Infrastructure Build-out to Power Artificial Intelligence Data Centers

ArXiv CS.AI 50%

arXiv:2609.08166v1 Announce Type: cross Abstract: The twenty-first century's transformative technology, artificial intelligence, is increasingly constrained by the twentieth century's transformative technology, the electricity grid. Rapid growth in electricity demand from data...

<https://arxiv.org/abs/2609.08166>

01 CONF

CALIPER: Clean Scenes Cannot Rank Physical Inference in Pretrained Visual Representations

ArXiv CS.AI 50%

arXiv:2609.08250v1 Announce Type: cross Abstract: How far a pushed object slides depends on its mass and friction, which no single image reveals. Pretrained visual encoders are increasingly used as the perception front end of world models for manipulation, and their physical...

<https://arxiv.org/abs/2609.08250>

02 CONF

ACEA: An Adversarial Co-Evolution Arena for Head-to-Head Red-Team and Blue-Team LLM Testing

ArXiv CS.AI 50%

arXiv:2609.08256v1 Announce Type: cross Abstract: Automated red-team attacks and blue-team defenses for large language models (LLMs) are advancing quickly. However, attackers and defenders are built and tested in isolation, and the resulting scores are hard to trust. To tackle...

<https://arxiv.org/abs/2609.08256>

03 CONF

Synergistic Fusion of Topological Structure and Temporal Semantics of Mobility for Urban Region Embedding

ArXiv CS.AI 50%

arXiv:2609.08268v1 Announce Type: cross Abstract: Urban region embeddings have shown promising results in diverse urban sensing tasks such as crime, income, and service-call prediction. Recent methods improve representation quality by integrating mobility data with auxiliary...

<https://arxiv.org/abs/2609.08268>

04 CONF

A Measurement Study of LLM Inference Trade-offs Across Edge Continuum Hardware

ArXiv CS.AI 50%

arXiv:2609.08307v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used as backends for intelligent web services, but serving them across the edge continuum requires balancing quality, latency, model footprint, and energy. This paper presents a...

<https://arxiv.org/abs/2609.08307>

05 CONF

Generating Pretraining Tokens from Organic Data for Data-Bound Scaling

ArXiv CS.AI 50%

arXiv:2605.17849v2 Announce Type: replace-cross Abstract: LLM pretraining is shifting from a compute-bound to a data-bound regime, where available human (organic) text falls far short of scaling demands. However, reaching the data-bound regime does not mean the model has fully...

<https://arxiv.org/abs/2605.17849>

06 CONF

Towards Embodied Air-Ground Cooperative Object Search: Benchmark, Dataset and Agentic Method

ArXiv CS.AI 50%

arXiv:2609.08402v1 Announce Type: cross Abstract: Air-Ground Object Search (AGOS) in urban environments is a challenging embodied task, which requires an Unmanned Aerial Vehicle (UAV) and an Unmanned Ground Vehicle (UGV) to jointly search for and verify a specified target...

<https://arxiv.org/abs/2609.08402>

07 CONF

Same Values, Different Languages? From Multilingual Probing to Steering LLMs Toward Chinese Social Values

ArXiv CS.AI 50%

arXiv:2609.08515v1 Announce Type: cross Abstract: As Large Language Models (LLMs) are increasingly integrated into human society, aligning them with pluralistic social values has become a critical priority. However, whether LLMs exhibit consistent value preferences across...

<https://arxiv.org/abs/2609.08515>

08 CONF

Let It Go or Learn to Self-Correct: Continuous Diffusion for Constrained Discrete Tasks

ArXiv CS.AI 50%

arXiv:2609.09009v1 Announce Type: cross Abstract: Denoising Diffusion Probabilistic Models (DDPMs) generate samples by starting from noise and repeatedly denoising while keeping each update close to the current noisy state. This behavior is effective in many continuous domains,...

<https://arxiv.org/abs/2609.09009>

09 CONF

The Unreliable Progress Bar: Can LLM Agents Reliably Report Task Progress Throughout Execution?

ArXiv CS.AI 50%

arXiv:2609.08589v1 Announce Type: cross Abstract: Recent large language models can emit task-progress signals that agent frameworks use to decide whether a task should continue or stop, yet whether a model can reliably report its task progress at every stage of a task, and where...

<https://arxiv.org/abs/2609.08589>

10 CONF

Neptune: An AI model for Global Ocean Subseasonal Prediction

ArXiv CS.AI 50%

arXiv:2609.08606v1 Announce Type: cross Abstract: Subseasonal-to-seasonal (S2S) forecasting is societally critical, supporting decision-making in sectors ranging from water and agricultural management to disaster risk reduction, energy planning, and insurance. Achieving reliable...

<https://arxiv.org/abs/2609.08606>

11 CONF

Leveraging contextual events on structure-aware next activity prediction

ArXiv CS.AI 50%

arXiv:2609.08622v1 Announce Type: cross Abstract: Predictive process monitoring aims at forecasting various aspects of running processes. Among the different tasks, next activity prediction represents the most extensively investigated. However, only a limited number of existing...

<https://arxiv.org/abs/2609.08622>

12 CONF

Difficulty-Adaptive Tree-Structured Policy Optimization for Expanding Reasoning Coverage in RLVR

ArXiv CS.AI 50%

arXiv:2609.08650v1 Announce Type: cross Abstract: Reinforcement Learning with Verifiable Rewards (RLVR) has been central to the recent success of Large Reasoning Models. However, while RLVR significantly improves single-sample accuracy, it often fails to expand the model's...

<https://arxiv.org/abs/2609.08650>

13 CONF

MoEMB: Scaling Universal Multimodal Embeddings with Efficient Mixture-of-Experts Models

ArXiv CS.AI 50%

arXiv:2609.08663v1 Announce Type: cross Abstract: Universal multimodal embedding (UME) increasingly demands encoder's capacity for handling a broad range of tasks and modalities with increased complexity. Prior scaling methods either increase the representation size, retrieval...

<https://arxiv.org/abs/2609.08663>

14 CONF

X2Streaming-ASR: wait when uncertain, emit when ready for streaming ASR

ArXiv CS.AI 50%

arXiv:2609.08672v1 Announce Type: cross Abstract: Streaming automatic speech recognition (ASR) for real-time voice agents and full-duplex dialogue must provide accurate partial transcripts with low commit latency. Existing systems commonly use a fixed chunk size, look-ahead, or...

<https://arxiv.org/abs/2609.08672>

15 CONF

BIFTA: Brain-Inspired Few-Shot Tactile Adaptation for Unknown Sensors

ArXiv CS.AI 50%

arXiv:2609.08673v1 Announce Type: cross Abstract: Advances in tactile sensing have made contact-rich perception possible, accelerating progress in robotic manipulation, material understanding, and embodied interaction. However, because optical design, elastomer mechanics, and...

<https://arxiv.org/abs/2609.08673>

16 CONF

Neither Adversarial Training Nor Purification: Emergent Adversarial Robustness from Oscillatory Predictive Learning

ArXiv CS.AI 50%

arXiv:2609.08683v1 Announce Type: cross Abstract: Adversarial robustness in computer vision is still largely achieved through adversarial training or test-time adversarial purification, both of which introduce significant computational overhead by generating adversarial examples...

<https://arxiv.org/abs/2609.08683>

17 CONF

SymbolicLight V1: Spike-Gated Dual-Path Language Modeling at High Encoder Spike Sparsity

ArXiv CS.AI 50%

arXiv:2605.21333v3 Announce Type: replace-cross Abstract: Natively trained spiking language models must preserve information across time while operating through sparse binary activations, a combination that has produced a persistent quality gap relative to dense Transformers. We...

<https://arxiv.org/abs/2605.21333>

18 CONF

Silent Revision: Measuring Undisclosed Change in the Safety Frameworks of Frontier AI Developers

ArXiv CS.AI 50%

arXiv:2609.08789v1 Announce Type: cross Abstract: Frontier AI developers publish safety frameworks that commit them to evidencing whether their models are dangerous. The European Union and California now treat these documents as instruments of accountability, and both already...

<https://arxiv.org/abs/2609.08789>

19 CONF

Evaluating and Improving Evidence-Grounded Fact-Checking in LLMs via Multi-Round Evidence Ablation

ArXiv CS.AI 50%

arXiv:2609.08943v1 Announce Type: cross Abstract: Automatic fact-checking systems assess the veracity of claims given evidence from relevant documents. Large Language Models (LLMs) have demonstrated strong performance in fact-checking due to their general reasoning capabilities....

<https://arxiv.org/abs/2609.08943>

20 CONF

SQLMorph: Query Mutation and Fine-Grained Metrics for Text-to-SQL Evaluation

ArXiv CS.AI 50%

arXiv:2609.08950v1 Announce Type: cross Abstract: Text-to-SQL systems translate natural language queries into executable SQL, democratizing access to structured data. Despite recent advances driven by large language models (LLMs), evaluation remains a major bottleneck: public...

<https://arxiv.org/abs/2609.08950>

21 CONF

Transformers as In-Context Samplers: From Closed-Form Diffusion to Estimation-Free Sampling

ArXiv CS.AI 50%

arXiv:2609.08981v1 Announce Type: cross Abstract: A growing body of work establishes that large language models are not mere statistical memorizers, but are capable of in-context learning: performing inference at test time using only examples provided in the prompt, without any...

<https://arxiv.org/abs/2609.08981>

22 CONF

Measuring LLM Sycophancy under Sustained Multi-Turn Pressure

ArXiv CS.AI 50%

arXiv:2609.09090v1 Announce Type: cross Abstract: Large language models (LLMs) may abandon correct positions when users push back, exhibiting a failure mode known as sycophancy. Existing evaluations typically use short, pre-specified conversations and may therefore miss failures...

<https://arxiv.org/abs/2609.09090>

23 CONF

NOAH: Learning the Full Patient Journey. A Longitudinal Multimodal Time-Aware Model for Representation and Forecasting

ArXiv CS.AI 50%

arXiv:2609.09140v1 Announce Type: cross Abstract: The digitization of healthcare has generated vast, longitudinal, and multimodal patient records over a lifetime, yet fully exploiting these data to represent and predict patient state trajectories remains a critical challenge....

<https://arxiv.org/abs/2609.09140>

24 CONF

Fewer yet critical: Reducing Redundant Token Dependencies for Transformer-based Time Series Forecasting

ArXiv CS.AI 50%

arXiv:2503.06867v2 Announce Type: replace Abstract: Time series forecasting (TSF) is important in real-world applications. Recently, Transformer-based methods have achieved strong performance by modeling token dependencies through attention mechanisms. However, existing methods...

<https://arxiv.org/abs/2503.06867>

25 CONF

From the Fluency Fallacy to the Micro-to-Macro Validity Gap: Opportunities and Pitfalls of LLMs in Social Simulation

ArXiv CS.AI 50%

arXiv:2507.19364v3 Announce Type: replace Abstract: The integration of Large Language Models (LLMs) into social simulation has generated considerable enthusiasm, but also raises substantial methodological and epistemological challenges. This critical review examines the use of...

<https://arxiv.org/abs/2507.19364>

26 CONF

ReST-RL: Reinforcing LLM Reasoning through Unified Self-Training and Value-Guided Search

ArXiv CS.AI 50%

arXiv:2508.19576v3 Announce Type: replace Abstract: With respect to improving the reasoning accuracy of LLMs, the representative reinforcement learning (RL) method - Group Relative Policy Optimization (GRPO) - has achieved critical success, yet it still suffers from the issue of...

<https://arxiv.org/abs/2508.19576>

27 CONF

The human-authorship halo: attribution bias in literary style evaluation by humans and AI

ArXiv CS.AI 50%

arXiv:2510.08831v2 Announce Type: replace Abstract: As AI writing tools become widespread, we need to understand how both humans and machines evaluate literary style, a domain where objective standards are elusive and judgments are inherently subjective. We conducted controlled...

<https://arxiv.org/abs/2510.08831>

28 **CONF****On the Context Sensitivity of LLM Moral Judgment**

ArXiv CS.AI 50%

arXiv:2603.23114v2 Announce Type: replace Abstract: A human's moral decision depends heavily on the context. Yet research on LLM morality has largely studied fixed scenarios. We address this gap by introducing Contextual MoralChoice, a dataset of moral dilemmas with systematic...

<https://arxiv.org/abs/2603.23114>

29 **CONF****Reducing Hallucinations in LLM-based Scientific Literature Analysis Using Peer Context Outlier Detection**

ArXiv CS.AI 50%

arXiv:2604.01461v2 Announce Type: replace Abstract: Reducing hallucinations in Large Language Models (LLMs) is essential for accurate data extraction from large text corpora. Current methods, like prompt engineering and chain-of-thought prompting, focus on individual documents...

<https://arxiv.org/abs/2604.01461>

30 **CONF****Towards Trustworthy Physical AI: From Theory to Practice Across Life Cycle**

ArXiv CS.AI 50%

arXiv:2607.22877v3 Announce Type: replace Abstract: Physical AI refers to AI systems that understand, reason about, and act in accordance with the physical world and its underlying laws, dynamics, and constraints. Unlike conventional AI systems, physical AI interacts...

<https://arxiv.org/abs/2607.22877>

31 **CONF****Behavioral Grammar: Detecting Adaptive Malware via Tiny Language Model Priors and Second-Order Temporal Analysis**

ArXiv CS.AI 50%

arXiv:2608.00745v2 Announce Type: replace Abstract: Modern endpoint detection systems face a fundamental tension: signature-based approaches are trivially evaded by polymorphic or adaptive threats, while heavy deep-learning models resist auditability and deployment at scale....

<https://arxiv.org/abs/2608.00745>

32 **CONF****Who Delegates to AI? Evidence from Agent Configurations in Github**

ArXiv CS.AI 50%

arXiv:2608.20425v2 Announce Type: replace Abstract: A growing body of literature measures the extent to which occupations are exposed to AI, yet existing measures capture where AI could perform tasks rather than whether workers have actually adopted it. We introduce a distinct...

<https://arxiv.org/abs/2608.20425>

33 **CONF****EvoGenUI-Bench: Evaluating LLMs as Multi-Turn Generative UI Assistants**

ArXiv CS.AI 50%

arXiv:2608.29387v2 Announce Type: replace Abstract: Large language models can generate interactive web interfaces, but reliable generative UI requires maintaining an executable artifact as user requests evolve. We introduce EvoGenUI-Bench, a benchmark for multi-turn interface...

<https://arxiv.org/abs/2608.29387>

34 CONF

A Theoretical Analysis of Provable Compositional Generalization in Neural Networks: A Necessary and Sufficient Condition

ArXiv CS.AI 50%

arXiv:2505.02627v2 Announce Type: replace-cross Abstract: Compositional generalization is the ability to systematically process novel combinations of known components is a hallmark of human intelligence; however, its theoretical foundation in neural...

<https://arxiv.org/abs/2505.02627>

35 CONF

Provable Pluralistic Alignment: Multi-Party RLHF under Offline Human Feedback

ArXiv CS.AI 50%

arXiv:2403.05006v2 Announce Type: replace-cross Abstract: Pluralistic alignment requires learning from feedback that reflects persistent and potentially conflicting stakeholder preferences while ultimately selecting a single collective policy. We study this problem in offline...

<https://arxiv.org/abs/2403.05006>

36 CONF

SAEs Can Improve Unlearning: Dynamic Sparse Autoencoder Guardrails for Precision Unlearning in LLMs

ArXiv CS.AI 50%

arXiv:2504.08192v2 Announce Type: replace-cross Abstract: Machine unlearning is a promising approach to improve LLM safety by removing unwanted knowledge from the model. However, prevailing gradient-based unlearning methods suffer from issues such as high computational costs,...

<https://arxiv.org/abs/2504.08192>

37 CONF

Clinician-Friendly Foundation Models for Ophthalmic Image Diagnostics without Fine-Tuning or Technical Barriers

ArXiv CS.AI 50%

arXiv:2504.15928v3 Announce Type: replace-cross Abstract: Artificial intelligence (AI) shows remarkable potential in medical imaging diagnostics, yet most current models require retraining when applied across different clinical settings, limiting their scalability. We developed...

<https://arxiv.org/abs/2504.15928>

38 CONF

Evaluating the Scalability and Adversarial Generalization of GRPO-Trained NLI Models

ArXiv CS.AI 50%

arXiv:2504.18376v3 Announce Type: replace-cross Abstract: Natural Language Inference (NLI) is a central task in natural language understanding with applications in fact-checking, question answering, and information retrieval. Despite its importance, current NLI systems heavily...

<https://arxiv.org/abs/2504.18376>

39 CONF

Beyond Retrieval: Joint Supervision and Multimodal Document Ranking for Textbook Question Answering

ArXiv CS.AI 50%

arXiv:2505.13520v2 Announce Type: replace-cross Abstract: Textbook question answering (TQA) is a complex task, requiring the interpretation of complex multimodal context. Although recent advances have improved overall performance, they often encounter difficulties in educational...

<https://arxiv.org/abs/2505.13520>

40 CONF

Adaptive Nonlinear Vector Autoregression: Robust Forecasting for Noisy Chaotic Time Series

ArXiv CS.AI 50%

arXiv:2507.08738v3 Announce Type: replace-cross Abstract: Nonlinear vector autoregression (NVAR) and reservoir computing (RC) have shown promise in forecasting chaotic dynamical systems, such as the Lorenz-63 model and El Nino-Southern Oscillation. However, their reliance on...

<https://arxiv.org/abs/2507.08738>

41 CONF

PiMRef: Deducing Ever-evolving Spear-phishing Emails with Knowledge Base Invariants

ArXiv CS.AI 50%

arXiv:2507.15393v2 Announce Type: replace-cross Abstract: Phishing email is a critical step in the cybercrime kill chain due to the high reachability of victims' email accounts and the low cost of launching phishing campaigns. This ever-evolving nature of phishing emails makes...

<https://arxiv.org/abs/2507.15393>

42 CONF

Introducing HALC: A general pipeline for the systematic and reliable construction of prompts for automated coding with LLMs in the computational social sciences

ArXiv CS.AI 50%

arXiv:2507.21831v2 Announce Type: replace-cross Abstract: LLMs are seeing widespread use for task automation, including automated coding in the social sciences. However, even though researchers have proposed different prompting strategies, their effectiveness varies across LLMs...

<https://arxiv.org/abs/2507.21831>

43 CONF

Experimental Analysis of Productive Interaction Strategy with ChatGPT: User Study on Function and Project-level Code Generation Tasks

ArXiv CS.AI 50%

arXiv:2508.04125v2 Announce Type: replace-cross Abstract: The application of Large Language Models (LLMs) is growing in the productive completion of Software Engineering tasks. Yet, studies investigating productive prompting techniques often employed a limited problem space,...

<https://arxiv.org/abs/2508.04125>

44 CONF

When Tools Hurt LLM Reasoning: State-Dependent Belief Revision under External Evidence

ArXiv CS.AI 50%

arXiv:2508.15754v2 Announce Type: replace-cross Abstract: Tool use is often assumed to monotonically improve reasoning, where external evidence is expected to help when relevant and be ignored when irrelevant. We show that this assumption fails in a state-dependent way. Across...

<https://arxiv.org/abs/2508.15754>

45 CONF

Fusing Sequence Motifs and Pan-Genomic Features: Antimicrobial Resistance Prediction using an Explainable Lightweight 1D CNN-XGBoost Ensemble

ArXiv CS.AI 50%

arXiv:2509.23552v2 Announce Type: replace-cross Abstract: Antimicrobial Resistance (AMR) is a rapidly escalating global health crisis. While genomic sequencing enables rapid prediction of resistance phenotypes, current computational methods have limitations. Standard machine...

<https://arxiv.org/abs/2509.23552>

46 **CONF****DC-Gen: Post-Training Diffusion Acceleration with Deeply Compressed Latent Space****ArXiv CS.AI** **50%**

arXiv:2509.25180v3 Announce Type: replace-cross Abstract: Existing text-to-image diffusion models excel at generating high-quality images, but face significant efficiency challenges when scaled to high resolutions, like 4K image generation. While previous research accelerates...

<https://arxiv.org/abs/2509.25180>

47 **CONF****BaNEL: Exploration Posteriors for Generative Modeling Using Only Negative Rewards****ArXiv CS.AI** **50%**

arXiv:2510.09596v2 Announce Type: replace-cross Abstract: Today's generative models thrive with large amounts of supervised data and informative reward functions characterizing the quality of the generation. They work under the assumptions that the supervised data provides...

<https://arxiv.org/abs/2510.09596>

48 **CONF****ConsistencyAI: A Benchmark to Assess LLMs' Factual Consistency When Responding to Different Demographic Groups****ArXiv CS.AI** **50%**

arXiv:2510.13852v3 Announce Type: replace-cross Abstract: Is an LLM telling you different facts than it's telling me? This paper introduces ConsistencyAI, an independent benchmark for measuring the factual consistency of large language models (LLMs) for different personas....

<https://arxiv.org/abs/2510.13852>

49 **CONF****Deep Active Inference with Diffusion Policy and Multiple Timescale World Model for Real-World Exploration and Navigation****ArXiv CS.AI** **50%**

arXiv:2510.23258v2 Announce Type: replace-cross Abstract: Autonomous robotic navigation in real-world environments requires exploration to acquire environmental information as well as goal-directed navigation in order to reach specified targets. Active inference (AIF) based on...

<https://arxiv.org/abs/2510.23258>

50 **CONF****Reinforcement Learning Improves Traversal of Parametric Knowledge in LLMs****ArXiv CS.AI** **50%**

arXiv:2511.05933v3 Announce Type: replace-cross Abstract: Reinforcement learning (RL) is often credited with improving reasoning at the expense of factual knowledge. We instead find that reasoning models outperform their instruction-tuned versions on factual recall by accessing...

<https://arxiv.org/abs/2511.05933>

51 **CONF****Who Laughs with Whom? Disentangling Influential Factors in Humor Preferences across User Clusters and LLMs****ArXiv CS.AI** **50%**

arXiv:2601.03103v2 Announce Type: replace-cross Abstract: Humor preferences vary widely across individuals and cultures, complicating the evaluation of humor using large language models (LLMs). In this study, we model heterogeneity in humor preferences in Oogiri, a Japanese...

<https://arxiv.org/abs/2601.03103>

52 CONF

GIFT: Reconciling Post-Training Objectives via Variational Finite-Temperature Gibbs Initialization

ArXiv CS.AI 50%

arXiv:2601.09233v3 Announce Type: replace-cross Abstract: The prevailing post-training paradigm for Large Reasoning Models (LRMs)---Supervised Fine-Tuning (SFT) followed by Reinforcement Learning (RL)-suffers from an intrinsic optimization mismatch: the rigid likelihood...

<https://arxiv.org/abs/2601.09233>

53 CONF

Safety boundary maintenance in consumer AI systems responding to pediatric health queries: a cross-platform benchmark evaluation under naturalistic and adversarially pressured conditions

ArXiv CS.AI 50%

arXiv:2601.09721v2 Announce Type: replace-cross Abstract: Consumer artificial intelligence chatbots are now accessed by hundreds of millions of users seeking health information, yet systematic evaluation of their safety boundary maintenance under real-world caregiver pressure...

<https://arxiv.org/abs/2601.09721>

54 CONF

Boosting LLM Reasoning via Human-Inspired Reward Shaping

ArXiv CS.AI 50%

arXiv:2602.04265v4 Announce Type: replace-cross Abstract: Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a promising paradigm for enhancing reasoning in Large Language Models (LLMs). However, existing reward formulations typically treat exploration and...

<https://arxiv.org/abs/2602.04265>

55 CONF

US-JEPA: A Joint Embedding Predictive Architecture for Ultrasound

ArXiv CS.AI 50%

arXiv:2602.19322v2 Announce Type: replace-cross Abstract: Ultrasound (US) imaging poses unique challenges for representation learning due to its inherently noisy acquisition process. The low signal-to-noise ratio and stochastic speckle patterns hinder standard self-supervised...

<https://arxiv.org/abs/2602.19322>

56 CONF

CTC-TTS: LLM-based dual-streaming text-to-speech with CTC alignment

ArXiv CS.AI 50%

arXiv:2602.19574v2 Announce Type: replace-cross Abstract: Large-language-model (LLM)-based text-to-speech (TTS) systems can generate natural speech, but most are not designed for low-latency dual-streaming synthesis. High-quality dual-streaming TTS depends on accurate...

<https://arxiv.org/abs/2602.19574>

57 CONF

Asking the Right Questions: Ontology-Grounded Interpretable Embeddings for Biomedical Text

ArXiv CS.AI 50%

arXiv:2603.01690v3 Announce Type: replace-cross Abstract: While dense biomedical embeddings achieve strong performance, their opaque dimensions limit transparency in biomedical NLP. Recent question-based interpretable embeddings represent text through binary answers to...

<https://arxiv.org/abs/2603.01690>

58 CONF

Spatiotemporal Heterogeneity of AI-Driven Traffic Flow Patterns and Land Use Interaction: A GeoAI-Based Analysis of Multimodal Urban Mobility

ArXiv CS.AI 50%

arXiv:2603.05581v2 Announce Type: replace-cross Abstract: Urban traffic flow is governed by the complex, nonlinear interaction between land use configuration and spatiotemporally heterogeneous mobility demand. Conventional global regression and time-series models cannot...

<https://arxiv.org/abs/2603.05581>

59 CONF

Adapting Technical-Service LLM Agents with Latent Logic Augmentation, Robust Noise Reduction, and Hybrid Reward Modeling

ArXiv CS.AI 50%

arXiv:2603.18074v2 Announce Type: replace-cross Abstract: Technical-service LLM agents are entering production workflows, where value depends on whether engineers adopt generated replies. Service tickets hide decision logic, contain noisy single-reference responses, and make...

<https://arxiv.org/abs/2603.18074>

60 CONF

Behind Harmful Compliance: Behavioral and Mechanistic Divergence Across LLM Jailbreaks

ArXiv CS.AI 50%

arXiv:2604.18510v2 Announce Type: replace-cross Abstract: Open-weight language models can be rendered unsafe through several parameter-level interventions, yet models with matched harmful compliance can exhibit fundamentally different failure modes. We compare harmful supervised...

<https://arxiv.org/abs/2604.18510>

61 CONF

EPC-3D-Diff: Equivariant Physics Consistent Conditional 3D Latent Diffusion for CBCT to CT Synthesis

ArXiv CS.AI 50%

arXiv:2605.20470v2 Announce Type: replace-cross Abstract: Cone-beam CT (CBCT) is routinely acquired during radiotherapy for patient setup, but its quantitative reliability is degraded by scatter, noise, and reconstruction artifacts, limiting Hounsfield Unit (HU) accuracy. We...

<https://arxiv.org/abs/2605.20470>

62 CONF

Clarify, Abstain or Answer? Strategising in Conversation with Belief-Augmented Generation

ArXiv CS.AI 50%

arXiv:2605.25831v2 Announce Type: replace-cross Abstract: Large language models (LLMs) define a distribution over text, which can be viewed as a probabilistic representation of uncertainty: sampling K responses yields a belief state - responses a model deems plausible. Existing...

<https://arxiv.org/abs/2605.25831>

63 CONF

Linear Separability of Activation Representations after Supervised Fine-Tuning on Incorrect Responses: A Study of Synthetic Dishonesty in Large Language Models

ArXiv CS.AI 50%

arXiv:2605.30381v2 Announce Type: replace-cross Abstract: When a language model is fine-tuned to produce systematically incorrect responses, does this training leave a structured, linearly recoverable trace in its internal activations? We study this question in a controlled...

<https://arxiv.org/abs/2605.30381>

64 CONF

AgentServeSim: Serving-System Simulation and Policy Search for LLM Agent Programs

ArXiv CS.AI 50%

arXiv:2606.09613v3 Announce Type: replace-cross Abstract: Large language model agents execute programs comprising multiple model turns interleaved with external tool calls. Their job completion time depends on how the serving system retains KV state across tool gaps, routes...

<https://arxiv.org/abs/2606.09613>

65 CONF

Field-level weak lensing cosmology with \$60\$ simulations using multifidelity simulation-based inference

ArXiv CS.AI 50%

arXiv:2606.23346v2 Announce Type: replace-cross Abstract: We perform a realistic KiDS-Legacy mock analysis with field-level neural compression and simulation-based inference using just 60 N -body simulations. The weak lensing shear field encodes substantially more cosmological...

<https://arxiv.org/abs/2606.23346>

66 CONF

Challenges and Recommendations for LLM-as-a-Judge in Multilingual Settings and for Low-Resource Languages

ArXiv CS.AI 50%

arXiv:2607.02235v2 Announce Type: replace-cross Abstract: LLM-as-a-Judge has become the dominant evaluation paradigm for many natural language generation tasks (albeit mostly in English) due to shortcomings of conventional metrics and high correlations with human judgment. There...

<https://arxiv.org/abs/2607.02235>

67 CONF

Parameter-Free Dynamic Regret under Heavy-Tailed Noise

ArXiv CS.AI 50%

arXiv:2607.27073v2 Announce Type: replace-cross Abstract: We study online convex optimization with stochastic gradient noise whose conditional p -th central moment is bounded by σ^p , for an unknown $p \in (1, 2]$. For losses with Lipschitz bound G on a domain of diameter...

<https://arxiv.org/abs/2607.27073>

68 CONF

A False Average: Chain-of-Thought Monitors Collapse Where They Are the Only Defense

ArXiv CS.AI 50%

arXiv:2608.00583v2 Announce Type: replace-cross Abstract: Chain-of-thought (CoT) monitoring is meant to catch the reward hacks that look clean in the actions and betray themselves only in the reasoning. We show that this is exactly where an adversary who controls the reasoning...

<https://arxiv.org/abs/2608.00583>

69 CONF

An AI-Based Decision-Support Pipeline for Day-Ahead Photovoltaic Forecasting

ArXiv CS.AI 50%

arXiv:2608.02088v2 Announce Type: replace-cross Abstract: Reliable photovoltaic (PV) forecasts can support low-carbon energy systems, but deployed sites may have only short and incomplete records. Physical and hybrid methods can be sensitive to weather inputs, calibration, and...

<https://arxiv.org/abs/2608.02088>

70 CONF

Fusing Perceptual Vision Experts with Multimodal Large Language Models for Explainable Plant Disease Diagnosis: From Benchmark Imagery to Real-World Robotic Field Validation

ArXiv CS.AI 50%

arXiv:2608.24934v2 Announce Type: replace-cross Abstract: Accurate field plant disease diagnosis requires reliable fusion of uncertain and conflicting perceptual evidence. We present the Hybrid Hierarchical Multi-Agent Framework (H^2MAF), combining decision-level fusion of...$

<https://arxiv.org/abs/2608.24934>

71 CONF

Generative Action-Chunk Sampling for Adaptive Stiffness Control in Physical Human-Robot Collaboration

ArXiv CS.AI 50%

arXiv:2608.25284v2 Announce Type: replace-cross Abstract: Physical human-robot collaboration requires a robot to provide assistance when human intention is clear while remaining compliant when several future motions are plausible. We present an adaptive stiffness framework based...

<https://arxiv.org/abs/2608.25284>

72 CONF

Unfolding Scientific Papers into Multi-Turn Generation Trajectories for Continued Pre-Training

ArXiv CS.AI 50%

arXiv:2608.25826v2 Announce Type: replace-cross Abstract: A recent line of synthetic-data work reconstructs the thinking behind existing text rather than rewriting the text itself, but it operates on short web passages, recovers only local thoughts, and leaves the structure of...

<https://arxiv.org/abs/2608.25826>

73 CONF

Actionable CBFI: Integrating Structural Decomposition and Causal Counterfactual Recourse for Tabular Machine Learning

ArXiv CS.AI 50%

arXiv:2608.27821v2 Announce Type: replace-cross Abstract: Explainable artificial intelligence (XAI) increasingly calls for actionable counterfactual recourse, yet current methodologies face challenges related to causal invalidity, excessive cognitive burden, and predictive...

<https://arxiv.org/abs/2608.27821>